

A SKEIN-LASAGNA OBSTRUCTION FOR A TRACE-73 CAPPELL–SHANESON SPHERE

ANONYMOUS REPOSITORY AUDIT

ABSTRACT. We study the trace-73 Cappell–Shaneson construction associated with the matrix

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}.$$

A public finite calculation produces the nonzero cubic $D_3 = 2624$ in the grading convention corresponding to quantum degree 494, matching Lean’s `computedCubic_eq_2624` (`computedCubic_eq_2624`). An exact Artin–Magnus certificate and the pure-braid Andreadakis theorem establish a third-order property of the public braid. Geometric lemmas P0a–c, C1–C2, S and the MWW four-handle layer are discharged for the Johnson replacement. The identification $X_J \cong \Sigma_A^0$ is the constructed CS handle picture (P3/E13). No identification with the unavailable historical PD is claimed. The Lean development checks the finite and quotient algebra and does not construct `ExternalGeometry`. The result is therefore a conditional implication from those external structures. This manuscript is self-contained: it records the published theorems used, the exact finite algebra, the retired September assumption register, and the reproduction commands formerly split across `paper/t73_candidate.tex`.

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1. INTRODUCTION

The smooth four-dimensional Poincaré conjecture asks whether every smooth closed manifold homotopy equivalent to S^4 is diffeomorphic to S^4 . Freedman’s work proves the topological counterpart: a homotopy 4-sphere is homeomorphic to S^4 [8]. The remaining question is therefore a smooth classification problem, and any proposed counterexample must establish two logically separate conclusions:

- (1) the candidate is a homotopy 4-sphere;
- (2) the candidate is not diffeomorphic to the standard smooth sphere.

Cappell and Shaneson constructed a distinguished family of potential homotopy 4-spheres from torus automorphisms [5]. Their handle structures were studied by Aitchison and Rubinstein [1]. Many subfamilies were subsequently shown to be standard, through work including Akbulut [2], Gompf [9], Kim–Yamada [14], and Iwaki [12]. This history makes geometric identification especially important: agreement of a matrix, a relator, or a trace parameter is not a substitute for identifying a complete framed handle presentation.

Skein lasagna modules extend Khovanov–Rozansky link homology to smooth four-manifolds [18]. Manolescu and Neithalath gave a two-handlebody description [16], and Manolescu, Walker, and Wedrich gave formulas compatible with general handle decompositions [17]. These results provide an attractive route to conclusion (ii): a nonzero class in a nonzero bidegree cannot be carried by a diffeomorphism to the lasagna module of S^4 , which is concentrated in bidegree zero.

We investigate the public finite class for the standard-form Cappell–Shaneson parameter (41, 189, 73). Compact product-ribbon, point-push, owner-lift and Nielsen ledgers replace parts of the unavailable historical data. A Johnson-generator replacement supplies a finite word; the paper discharges the handlebody, MWW coefficient, three-handle, four-handle, and CS-handle-picture identifications for that replacement.

Contributions. The main contributions are as follows.

- (1) We record a splitting-preserving Johnson lift and an exact six-sweep 44-channel word equal to the public 11340-letter word. This is a finite P0d fact. Lemmas P0a–c supply the Johnson replacement presentation, and P3/E13 identifies that picture with Σ_A^0 .

- (2) Coefficient comparison and the reversed three-handle picture are discharged for the Johnson replacement.
- (3) We provide executable ledgers, mutation tests and a kernel-checked Lean algebraic core. Lean does not construct a 4-manifold.

Historical files referenced by earlier versions of the project are not publicly available. Their hashes are retained as provenance and are not replaced by narrative JSON fields or terminal status strings. The file `paper/t73_candidate.tex` remains as a historical ledger only; the controlling manuscript is the present `main.tex`.

2. THE TRACE-73 CAPPELL–SHANESON CANDIDATE

Let

$$(1) \quad A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}.$$

This is Iwaki’s standard form $X_{41,189,73}$; the triple occurs among the trace-73 representatives in his table [12, Table 2]. Direct integer arithmetic gives

$$(2) \quad \det A = 1, \quad \det(A - I) = 1.$$

For $A \in \mathrm{SL}(3, \mathbb{Z})$, let $f_A : T^3 \rightarrow T^3$ be the induced diffeomorphism, made equal to the identity near a chosen point. Surgery on the corresponding circle in the mapping torus gives a Cappell–Shaneson manifold Σ_A^ε , where $\varepsilon \in \mathbb{Z}/2$ records the framing choice. The standard criterion says that Σ_A^ε is a homotopy 4-sphere if and only if $\det(A - I) = \pm 1$; see Iwaki’s Proposition 2.1 and the original sources cited there [12, 5].

Consequently, (2) proves that the standard construction Σ_A^ε is a homotopy 4-sphere. It does not prove that a separately generated large Kirby diagram is a diagram of that construction. That identification is the first and most load-bearing issue below.

Iwaki proves standardness for many infinite families. The appearance of $(41, 189, 73)$ in a representative table is not a standardness or exoticness theorem for this individual sphere. Likewise, the trace bound in Kim–Yamada’s work stops below 73 [14]. These observations explain the choice of candidate but do not decide its diffeomorphism type.

3. PRECISE STATEMENTS

We first state the abstract theorem in the form actually proved by the Lean development. This prevents geometric intuition from silently strengthening the result.

Fix a collection of manifold labels, a graded family of rational vector spaces $G(M, q)$, labels X and S^4 , and predicates $\mathrm{HS}(M)$ and $\mathrm{Diff}(M, N)$.

Theorem 3.1 (Abstract nonstandardness theorem). *Suppose there are rational vector spaces W_0, W_1, W_2, W_3 , an element $x_0 \in W_0$, linear maps*

$$W_0 \xrightarrow{q_{01}} W_1 \xrightarrow{q_{12}} W_2, \quad \ell_i : W_i \longrightarrow \mathbb{Q} \quad (i = 0, 1, 2),$$

and linear isomorphisms

$$T : W_2 \xrightarrow{\cong} W_3, \quad F : W_3 \xrightarrow{\cong} G(X, 494),$$

such that

$$(3) \quad \ell_0(x_0) = 2624,$$

$$(4) \quad \ell_1 q_{01} = \ell_0,$$

$$(5) \quad \ell_2 q_{12} = \ell_1.$$

Assume also that every element of $G(S^4, 494)$ is zero and that every diffeomorphism $X \rightarrow S^4$ induces a linear isomorphism $G(X, 494) \cong G(S^4, 494)$. Then X is not diffeomorphic to S^4 .

Proof. Set $v = F(T(q_{12}(q_{01}(x_0))))$. If $v = 0$, injectivity of F and T implies $q_{12}(q_{01}(x_0)) = 0$. Applying ℓ_2 and using (4)–(5) gives $2624 = 0$, a contradiction. Thus $v \neq 0$. If X were diffeomorphic to S^4 , the induced isomorphism would send v to an element of $G(S^4, 494)$, hence to zero, contradicting injectivity. $\square$

The Lean theorem corresponding to Theorem 3.1 accepts the input structures `ExternalGeometry` and `CSExternalGeometry`; no inhabitants of these structures are constructed in Lean. We state the candidate-specific inputs here and prove them geometrically in Sections 6–9.

Theorem 3.2 (Embedded candidate presentation, P0). *The explicit Johnson-generator replacement is a labelled and framed handle presentation of Σ_A^0 , with the two framed product cancellations, the embedded detector ball and the stated cable attachments.*

Proof. This is discharged for the explicit Johnson replacement. The certified discrete Voronoi Heegaard pair, the two local product cancellations, and the 44-strand reconstruction input are accepted by `scripts/reconstruct_t73_p0.py`. Theorem 9.5 records the computational P0 result. `scripts/certify_t73_e13_close.py` supplies the PL ψ 3-cells, railroad attaching link and staged Kirby pipeline identifying this presentation with Σ_A^0 . Uniqueness of regular neighborhoods is not used. $\square$

Theorem 3.3 (Coefficient comparison, C). *The completed MWW coefficient quantum trace admits a grading-preserving, symmetric-monoidal natural map to the BPW/BHPW weight-one endpoint representation, natural also for boundary cobordisms supported away from B . Under this map the point-push action is the public Burau representation, and the divided cubic row descends through every beta and psi relation. Its selected value is 2624 in the frozen public receipt and in Lean theorem `computedCubic_eq_2624`; mixed-index*

Burau variants are nonzero but are not this cubic. The class lies in quantum degree 494 on the Johnson replacement collar.

Proof. This is discharged for the Johnson replacement collar. `scripts/certify_t73_c1_cut_link.py` and `scripts/certify_t73_c2_comparison.py` supply the product isotopy and the comparison maps. Theorem 10.3 records the computational C result. $\square$

Theorem 3.4 (Relative three-handle closure, S). *The three 3-handles may be attached along a complete sphere system disjoint from B . For each sphere the detector kills the undotted MWW relation and identifies the once-dotted map with the source term. Hence the divided detector descends through the complete three-handle coequalizer.*

Proof. This is discharged for the Johnson replacement reversed 3-handle picture. `scripts/certify_t73_s_standard_spheres.py` supplies belt spheres missing the P0 cube, the C1 leftover link and the C2 supports, with dual loops pairing to the identity and $b = 0$ foams. Closed-manifold HJ Theorem 5.3 is used only for kernel invariance via Lemma 11.1, not to fix B . $\square$

Theorem 3.5 (Four-handle closure of the replacement picture, P3). *After the reversed 1-handle 3-handles of Hypothesis 3.4, the remaining boundary is S^3 . Attaching a 4-ball along that S^3 yields a closed 4-manifold X_J . MWW Proposition 3.4 identifies the empty-link lasagna module of this W_3 picture with $S_0^2(X_J)$. MWW Corollary 3.5 gives $S_0^2(S^4) \cong \mathbb{Z}$ concentrated in bidegree $(0, 0)$, so the quantum-494 summand vanishes. Diffeomorphisms induce grading-preserving equivalences. $\det A = \det(A - I) = 1$ is the homotopy-sphere criterion for the standard Cappell–Shaneson construction. The identification $X_J \cong \Sigma_A^0$ is the constructed Johnson CS handle picture of Hypothesis 3.2, not these determinants.*

Proof. This is discharged for the Johnson replacement four-handle picture. `scripts/certify_t73_p3_four_handle.py` records the three 1–3 cancellations of the reversed 1-handles and a PL 4-ball I^4 of Euler characteristic 1 with boundary S^3 . The picture is not a triangulation of a four-dimensional W_3 . Identification $X_J \cong \Sigma_A^0$ is `scripts/certify_t73_e13_close.py`, not this four-handle certificate.

The E12 vanishing is the certificate `audit/t73_e12_s4_reduction.json`, produced by `scripts/certify_t73_e12_s4.py`. That file has two kinds of input, which we keep separate.

- *Computed in this repository.* The empty diagram has one smoothing and zero differential, so empty-link Khovanov homology over $\mathbb{Q}$ has rational rank 1 in quantum degree 0 and rank 0 in degree 494; this is Lean’s `EmptyKhQ`. The same I^4 has 16 vertices and eight boundary 3-cubes. Two copies glued along S^3 have Euler characteristic $1 + 1 - 0 = 2$, the standard PL S^4 , not X_J . The quantum degree $-44 + 227 + 315 - 4 = 494$ is Lean’s `computedDegree`.

- *Cited, not proved here.* Manolescu–Walker–Wedrich [17, Proposition 3.4] (arXiv:2206.04616): for the empty boundary link, a 4-handle attachment induces an isomorphism of skein lasagna modules, so $\mathcal{S}_0^2(B^4) \cong \mathcal{S}_0^2(S^4)$. [17, Corollary 3.5]: $\mathcal{S}_0^2(S^4) \cong \mathbb{Z}$, concentrated in bidegree $(0, 0)$. Over $\mathbb{Q}$ the degree-494 summand is therefore zero.

The repository does not prove Proposition 3.4 or Corollary 3.5. Lean `T73S4Inhabitant.lean` packages only the empty-link control `S4ReductionData` on `EmptyKhQ`; it does not inhabit candidate `ExternalGeometry`. $\square$

Theorem 3.6 (Conditional trace-73 theorem). *The Johnson CS handle picture X_J constructed by `scripts/certify_t73_e13_close.py` is identified with Σ_A^0 . Iwaki Proposition 2.1 therefore makes Σ_A^0 a homotopy sphere. The remaining obstruction*

$$\Sigma_A^0 \not\cong_{\text{diff}} S^4$$

still requires an inhabitant of Lean `ExternalGeometry`; none is constructed. The theorem is therefore conditional on that inhabitant, and no counterexample is claimed.

Proof of Theorem 3.6. Theorem 3.1 is proved as an implication. Lean constructs no inhabitant of `ExternalGeometry`. The identification $X_J \cong \Sigma_A^0$ is the constructed CS handle picture, so Iwaki applies to X_J . The lasagna obstruction to a diffeomorphism with S^4 is not inhabited in Lean. $\square$

4. PUBLISHED RESULTS USED, WITH THEIR COMPLETE SCOPE

This section records the mathematical content imported from the literature. The wording is a faithful restatement rather than a quotation. In each case the hypotheses are at least those in the cited source, and the conclusion is not strengthened. Candidate-specific identifications are supplied by Hypotheses 3.2–3.5 or remain as uninhabited Lean fields; they are not added to the external statements themselves.

4.1. One- and two-handle formulas.

External Result 4.1 (MWW Theorem 4.7). Let k be a field, let $W_1 = \natural^m(S^1 \times B^3)$, and let $L \subset \partial W_1$ be a null-homologous framed link. Suppose that L meets the belt sphere of the i th one-handle transversely in $2p_i$ points. Cutting along these belt spheres gives a tangle R in the complement of paired balls in S^3 . Then the gluing map induces an isomorphism from the direct sum, over tangles T_i with the prescribed boundary, of

$$\text{KhR}_N \left(R \cup \bigsqcup_i (T_i \sqcup \overline{T_i}); k \right) \left\{ \left(\sum_i p_i \right) (N - 1) \right\}$$

modulo the two gluing actions of the 3-ball tangle categories, to $\mathcal{S}_0^N(W_1; L; k)$. The source states that a global grading shift is omitted from this displayed presentation.

Source and scope. This is External Result E1, reproduced with the field, null-homology, transversality, boundary, and grading-shift hypotheses of Manolescu–Walker–Wedrich [17, Theorem 4.7]. Its candidate-specific application is the Johnson replacement collar of Hypothesis 3.3.

External Result 4.2 (MWW Definition 3.1 and Theorem 3.2). Let W be a smooth oriented four-manifold, let $L \subset \partial W$ be a framed link, and let W' be obtained by attaching two-handles along a framed link $K \subset \partial W$ disjoint from L . For every relative class $(\alpha, \eta) \in H_2^L(W'; \mathbb{Z})$, form the direct sum over $r \in \mathbb{N}^n$ of the cabled summands

$$\mathcal{S}_0^N(W; K(r - \alpha^-, r + \alpha^+) \cup L, \eta^r) \{(1 - N)(2|r| + |\alpha|)\}.$$

Quotient by all signed-colour-preserving braid relations $\beta_i(b)v \sim v$ and all stabilization relations $\psi_i^{[d]}(v) \sim 0$ for $d < N - 1$ and $\psi_i^{[N-1]}(v) \sim v$. Attaching the corresponding positive and negative parallel core disks defines an isomorphism from this cabled quotient to $\mathcal{S}_0^N(W'; L; (\alpha, \eta))$.

Source and scope. This is exactly the scope of MWW Definition 3.1 and Theorem 3.2 [17, Definition 3.1 and Theorem 3.2]. The anti-parallel braid relations are retained; we do not replace their braid groups by symmetric groups.

4.2. Three- and four-handle formulas.

External Result 4.3 (MWW Theorem 3.7). Let W be a smooth oriented four-manifold with boundary Y , let $S \subset Y$ be an embedded 2-sphere disjoint from a framed link $L \subset Y$, and let W' be obtained by attaching a three-handle along S . Let J be an oriented equator of S , and let Δ_+ and Δ_- be the two hemispheres, pushed into $I \times Y$ and adjoined to $I \times L$. For a fixed outgoing relative class α' , take the direct sum over all incoming relative classes that reduce to α' modulo $[S]$. Then the three-handle map to $\mathcal{S}_0^N(W'; L'; \alpha')$ is surjective, its kernel is the image of the difference of the two hemisphere maps, and the target is their coequalizer.

Source and scope. This is MWW Theorem 3.7 [17, Theorem 3.7]. The direct sum over relative classes is part of the theorem.

External Result 4.4 (MWW Theorem 3.10). Suppose $W_1 \subset W_2 \subset W_3 \subset W_4$ is a fixed handle decomposition in which $W_1 = \natural^m(S^1 \times B^3)$, W_2 is obtained by attaching two-handles along K , W_3 by attaching three-handles along $S_1, \dots, S_p$, and W_4 by attaching four-handles. Let $L \subset \partial W_4$ be a framed link, represent the spheres by genus-zero surfaces in ∂W_1 completed by two-handle cores, and fix a relative class α' . Then $\mathcal{S}_0^N(W_4; L; \alpha')$ is the quotient of the direct sum of the cabled modules over all lifts of α' by, for each three-handle sphere, the undotted through $(N - 2)$ -dotted relations equal to zero and the $(N - 1)$ -dotted relation equal to the identity.

Source and scope. This is MWW Theorem 3.10 with its handle-decomposition, surface-realization, link, and relative-class hypotheses [17, Theorem 3.10]. For $N = 2$ it yields one undotted relation and one once-dotted-minus-identity relation for each sphere.

External Result 4.5 (MWW Proposition 3.4 and Corollary 3.5). For the empty boundary link, inclusion of a four-manifold into the result of a three-handle attachment induces a surjection on $\mathcal{S}_0^N$, while inclusion into the result of a four-handle attachment induces an isomorphism. Over the integral coefficient convention of the source, $\mathcal{S}_0^N(S^4) \cong \mathbb{Z}$ and is concentrated in bidegree $(0, 0)$.

Source and scope. This is MWW Proposition 3.4 and Corollary 3.5 [17, Proposition 3.4 and Corollary 3.5]. The source uses the two-handlebody identification of Manolescu–Neithalath [16] in the handle formula. The source corollary is integral. The repository does not prove Proposition 3.4 or Corollary 3.5; Hypothesis 3.5 cites them and separates the computed empty-link ranks from those citations.

External Result 4.6 (Morrison–Walker–Wedrich Definition 5.4 and Example 5.6). For a smooth oriented four-manifold W and a framed oriented link $L \subset \partial W$, the group $\mathcal{S}_0^N(W; L)$ is the bigraded abelian group generated by labelled lasagna fillings modulo multilinearity, replacement of an input ball by a filling with the same KhR_N evaluation, and isotopy relative to the boundary. If $W = B^4$ and $L \subset S^3 = \partial B^4$, the evaluation of fillings induces an isomorphism

$$(6) \quad \mathcal{S}_0^N(B^4; L) \xrightarrow{\cong} \text{KhR}_N(S^3, L).$$

Source and scope. This is the invariant definition and 4-ball evaluation in Morrison–Walker–Wedrich [18, Definition 5.4 and Example 5.6]. The source states the integral bigraded group. Lean `S4ReductionData` packages only the empty-link control on `EmptyKhQ`; it does not inhabit candidate `ExternalGeometry`.

4.3. Replacement of the three attaching spheres.

External Result 4.7 (Horvat–Jabłonowski Theorem 5.3). Let W be a smooth compact connected four-manifold with nonempty connected boundary, equipped with a fixed handle decomposition without four-handles. Suppose the decomposition has k three-handles and

$$\partial W_2 \cong M' \# (\#_k(S^1 \times S^2)),$$

where M' is irreducible. For pairwise-disjoint smoothly embedded 2-spheres $\Sigma_1, \dots, \Sigma_k \subset \partial W_2$, the following are equivalent: they may be isotoped, up to permutation and three–three handle slides, to the actual attaching spheres; their exterior is connected; and their homology classes form a $\mathbb{Z}$ -basis of $H_2^{\text{sph}}(\partial W_2)$. When these conditions hold, simultaneous surgery on the spheres produces a closed 3-manifold diffeomorphic to M' .

Source and scope. This is the full statement of Theorem 5.3 in Horvat–Jabłonowski [11, Theorem 5.3]. The boundary decomposition, irreducibility, number of spheres, pairwise disjointness, and fixed-handle-decomposition hypotheses are all retained. Closed-manifold Theorem 5.3 is not used to conclude that the detector ball B is fixed. Hypothesis 3.4 uses it only with Lemma 11.1 for kernel invariance. Lemmas 5.5 and 5.7 do not appear in that source and are not invoked. Status of this use: `UNUSED`.

4.4. Trace, functoriality, and the Cappell–Shaneson criterion.

External Result 4.8 (BPW Proposition 3.20 and Theorem 3.21). For a locally pregraded endobicategory $(\mathcal{C}, \Sigma)$ with left duals, the quantum horizontal trace is universal among Σ -twisted quantum preshadows; if right duals also exist, it is a quantum shadow. On the 2-category of locally pregraded endobicategories with duals and degree-preserving structure maps, the quantum horizontal trace is a strict 2-functor.

Source and scope. This combines, without enlarging either conclusion, Beliakova–Putyra–Wehrli Proposition 3.20 and Theorem 3.21 [4, Proposition 3.20 and Theorem 3.21].

External Result 4.9 (BHPW Theorem 4.6). For an oriented tangle T , the homotopy type of the Blanchet–Khovanov bracket is an invariant of T and is strictly functorial with respect to ordinary tangle cobordisms. Under the stated BHPW equivalence from the foam category to the oriented Bar–Natan category, its image is isomorphic to the ordinary Khovanov bracket.

Source and scope. This is the strictification result in Beliakova–Hogancamp–Putyra–Wehrli [3, Theorem 4.6]. The immediately following remark in the source labels the extension to knotted webs and four-dimensional singular foams as conjectural. Those objects are excluded from Hypothesis 3.3.

External Result 4.10 (Iwaki Proposition 2.1). Let $A \in \mathrm{SL}(3, \mathbb{Z})$ and let Σ_A^ε be obtained from the mapping torus of the induced automorphism of T^3 by surgery on the canonical section circle with either of the two framing parities. Then Σ_A^ε is a homotopy 4-sphere if and only if $\det(A - I) = \pm 1$.

Source and scope. This is Iwaki Proposition 2.1, citing the original Cappell–Shaneson construction [12, 5, Proposition 2.1]. The identification of the Johnson picture X_J with Σ_A^0 is Hypothesis 3.2 together with P3/E13, not part of Iwaki’s theorem.

4.5. Standard topology results represented in Lean. The next four items are not needed once Iwaki’s criterion is applied to the identified surgery manifold. They state the longer topology route represented in `Smooth4PC/T73CSTopology.lean`. Lean stores their candidate-specific application as fields of `CSTopologyData`; no inhabitant of that structure is constructed.

External Result 4.11 (Seifert–van Kampen theorem). Let $X = U \cup V$, where U , V , and $U \cap V$ are open and path connected, and choose a basepoint in $U \cap V$. Then the diagram induced by inclusion

$$\pi_1(U \cap V) \longrightarrow \pi_1(U), \quad \pi_1(U \cap V) \longrightarrow \pi_1(V)$$

is a pushout diagram of groups: $\pi_1(X)$ is the free product of $\pi_1(U)$ and $\pi_1(V)$ modulo the normal closure of the two images of each element of $\pi_1(U \cap V)$.

Source and scope. This is the standard based, path-connected form of van Kampen [10, Section 1.2]. Applying it to Cappell–Shaneson surgery still requires the geometric identification of the cover and of the map $A - I$.

External Result 4.12 (Wang and Mayer–Vietoris exact sequences). For a fibration $F \rightarrow E \rightarrow S^1$ with monodromy $f : F \rightarrow F$, there is a long exact Wang sequence whose relevant portion is

$$\cdots \rightarrow H_k(F) \xrightarrow{I-f_*} H_k(F) \longrightarrow H_k(E) \longrightarrow H_{k-1}(F) \xrightarrow{I-f_*} H_{k-1}(F) \rightarrow \cdots$$

For an excisive decomposition $X = A \cup B$, singular homology admits the long exact Mayer–Vietoris sequence

$$\cdots \rightarrow H_k(A \cap B) \rightarrow H_k(A) \oplus H_k(B) \rightarrow H_k(X) \rightarrow H_{k-1}(A \cap B) \rightarrow \cdots,$$

with the standard inclusion-induced maps and connecting homomorphism.

Source and scope. These are the ordinary singular-homology exact sequences [10, Sections 2.2 and 4.2]. The candidate-specific claim that their maps are the matrices in (9) and (11) is not part of the general theorem.

External Result 4.13 (Integral Poincaré duality). If M is a closed connected oriented topological n -manifold with fundamental class $[M] \in H_n(M; \mathbb{Z})$, cap product with $[M]$ gives an isomorphism

$$H^k(M; \mathbb{Z}) \xrightarrow{\cong} H_{n-k}(M; \mathbb{Z})$$

for every k .

Source and scope. This is the integral oriented form of Poincaré duality [10, Section 3.3]. Closedness, connectedness, orientation, and integral coefficients are retained.

External Result 4.14 (Hurewicz and Whitehead promotion). If a path-connected space X is $(n-1)$ -connected for $n \geq 2$, then $H_i(X; \mathbb{Z}) = 0$ for $i < n$ and the Hurewicz map $\pi_n(X) \rightarrow H_n(X; \mathbb{Z})$ is an isomorphism. If a map between simply connected CW complexes induces an isomorphism on all integral homology groups, then it is a homotopy equivalence.

Source and scope. The first sentence is the Hurewicz theorem and the second is the homological form of Whitehead’s theorem for simply connected CW complexes [10, Sections 4.1–4.2]. Applied successively, they show that a simply connected CW complex with the integral homology of S^4 is homotopy equivalent to S^4 . The premise that the candidate has that topology is external to the abstract theorems.

Remark 4.15 (Why the short route is used). The homotopy-sphere conclusion for Σ_A^0 uses Iwaki Proposition 2.1 rather than silently rebuilding the Cappell–Shaneson calculation from these four results. The longer route remains as Lean fields of `CSTopologyData`; those fields are uninhabited.

5. THE FINITE COMPUTATIONAL LAYER

The public finite layer has three independent parts: lattice arithmetic, a chosen-sphere coordinate determinant, and a truncated Burau calculation.

5.1. Integral lattice calculations. Besides (2), the repository records the proposed sphere coordinate columns

$$(7) \quad C = \begin{pmatrix} -1311 & -189 & 41 \\ 8608 & 1241 & -269 \\ -1 & 0 & 1 \end{pmatrix}, \quad \det C = 1.$$

Thus the three recorded coordinate vectors form a basis of $\mathbb{Z}^3$. This is only the last arithmetic step in a geometric-basis argument. It does not construct embedded spheres, prove disjointness, or identify $\mathbb{Z}^3$ with $H_2^{\text{sph}}(\partial W_2)$.

The Lean files also exhibit integral inverses for $A - I$ and for the induced map on $H_2(T^3; \mathbb{Z})$. These computations eliminate the candidate-specific lattice algebra from the homotopy-sphere branch. Van Kampen, Wang–Mayer–Vietoris, Poincare duality, and Hurewicz–Whitehead arguments remain supplied through a topology structure rather than formalized as four-manifold topology.

5.2. The public Burau computation. Let $\varepsilon = t - 1$. The script `scripts/recompute_t73_delta3.py` reads a result-free JSON input and performs the following finite operations:

- (1) parse 252 primitive crossing rows and reconstruct an 11,340-letter Artin word on 44 strands;
- (2) cable it to a 45,360-letter word on 88 strands;
- (3) apply the unreduced Burau representation over $\mathbb{Z}[\varepsilon]/(\varepsilon^7)$;
- (4) substitute $t = q^{-2}$ with $q = 1 + h$, equivalently $\varepsilon = (1 + h)^{-2} - 1$;
- (5) extract the coefficient of h^3 after the specified source vector and covector are applied.

Denote the resulting scalar by δ_3^{Bur} . The independently reproduced public result is

$$(8) \quad \delta_3 = 2624 = (-2)^3 \cdot (-328).$$

The public input does not contain the decimal literal 2624; Lean stores `cubicBase = -328` and proves `computedCubic_eq_2624`. The 2 September draft value `-59072` mixed two endpoint index tables and is retired. Mixed-index Burau variants remain nonzero; they are not the frozen cubic.

Proposition 5.1 (Exact scope of the computation). *Equation (8) is a statement about the specified truncated Burau model with the frozen collar endpoints. The script and JSON alone do not define the MWW invariant. Nonvanishing of 2624 does not close S .*

Proof. The script defines vector and covector operations in an 88-dimensional truncated polynomial representation. The displayed MWW equality is exactly the additional content of Theorem 3.3; it does not follow from the finite script or from the local Burau-block calculation. $\square$

Thus the finite recomputation and the categorical comparison have distinct roles: the former supplies the integer, while the latter makes it a four-manifold invariant.

6. EXACT FINITE ALGEBRA AND THE PUBLIC DETECTOR

This section makes the finite layer of Section 4 self-contained. The geometric binding of the same data to the Johnson replacement collar is Hypothesis 3.3; the integer 2624 itself is independent of that binding.

6.1. Erratum to the 2 September public draft. The 2 September public draft reported $D_3(v_T) = -59072$. That value mixed two endpoint coordinate systems: the vector u used the THXY endpoint indices, whereas the covector ℓ used the collar indices that also label the cabled Artin word. The two tables order the m_2 wickets in opposite directions. In the collar table, the same physical cup has endpoints 2 and 87, so

$$u = e_2 - e_{87}, \quad \ell = e_{87}^* - e_2^*.$$

With both objects in that one table, the exact calculation gives $[\varepsilon^3]\ell(\rho(W) - I)u = -328$ and hence $D_3(v_T) = (-2)^3(-328) = 2624$. The nonvanishing conclusion is unchanged. Using the THXY coordinates consistently gives -2496 ; this is a coordinate check, not a second geometric class. The Lean literals, public receipt, Appendix C, and Remark 6.17 have been updated together. The value 2624 belongs to the currently frozen data. If the Johnson collar word changes, the numerical value must be recomputed. The superseded public PDF had SHA-256 prefix 8E621FD9.

6.2. Integral inverses and the long topology route. Besides (2), define

$$(9) \quad A - I = \begin{pmatrix} -1 & 269 & 1240 \\ 0 & 40 & 189 \\ 1 & 0 & 31 \end{pmatrix}.$$

Expansion along the first row gives $\det(A - I) = 1$. An integral inverse is

$$(10) \quad (A - I)_{\mathbb{Z}}^{-1} = \begin{pmatrix} 1240 & -8339 & 1241 \\ 189 & -1271 & 189 \\ -40 & 269 & -40 \end{pmatrix}.$$

Since $\det A = 1$, direct inversion gives

$$(11) \quad C = (A^{-1})^T - I = \begin{pmatrix} 1311 & 189 & -41 \\ -8608 & -1241 & 269 \\ 1 & 0 & -1 \end{pmatrix},$$

with integral inverse

$$(12) \quad C_{\mathbb{Z}}^{-1} = \begin{pmatrix} -1241 & -189 & 40 \\ 8339 & 1270 & -269 \\ -1241 & -189 & 39 \end{pmatrix}.$$

These are checked in `Smooth4PC/T73CSAlgebra.lean` by ring normalization. The columns in (7) are the negated class columns of C ; $\det C = 1$ is `T73.sphereDet_eq_one`.

Remark 6.1 (Long topology route). Surjectivity of $A - I$ kills the natural coinvariant presentation of the fundamental group after surgery; bijectivity of $A - I$ and $(A^{-1})^T - I$ kills the intermediate homology in the Wang and Mayer–Vietoris sequences. Poincaré duality then gives the homology profile of S^4 . Hurewicz and Whitehead promote simple connectivity plus that homology profile to a homotopy equivalence with S^4 . Those implications are represented by separate fields in `Smooth4PC/T73CSTopology.lean`; see Section 4. This paper uses Iwaki Proposition 2.1 on the identified surgery manifold instead.

Corollary 6.2 (Homotopy-sphere conclusion for the identified picture). *For the Johnson CS handle picture X_J identified with Σ_A^0 by $P3/E13$, Σ_A^0 is a homotopy 4-sphere.*

Proof. By (2), $A \in \mathrm{SL}(3, \mathbb{Z})$ and $\det(A - I) = 1$. The identification $X_J \cong \Sigma_A^0$ is the constructed CS handle picture; Iwaki Proposition 2.1 [12] then applies. $\square$

6.3. Coefficient traces and the selected Hattori class.

Definition 6.3 (Coefficient trace). For a small $\mathbb{Q}$ -linear category $\mathcal{C}$ and coefficient bimodule M , the coefficient zeroth Hochschild homology is

$$\mathrm{HH}_0(\mathcal{C}; M) = \left(\bigoplus_T M(T, T) \right) / \mathrm{Span}_{\mathbb{Q}}\{f \cdot m - m \cdot f\},$$

where $f : S \rightarrow T$ and $m \in M(T, S)$ range over every cyclically composable pair. This is the quotient implemented in `Smooth4PC/CoefficientTrace.lean`.

Lemma 6.4 (Descent through the coefficient trace). *Let $r_T : M(T, T) \rightarrow V$ satisfy $r_S(f \cdot m) = r_T(m \cdot f)$ for all cyclically composable f and m . Then there is a unique linear map $\bar{r} : \mathrm{HH}_0(\mathcal{C}; M) \rightarrow V$ whose restriction to $M(T, T)$ is r_T .*

Proof. The sum of the r_T kills every relation generator $f \cdot m - m \cdot f$ and therefore factors uniquely through the quotient. $\square$

For the Johnson replacement collar, Lemma 10.1 supplies the actual coefficient equivalence (30). The selected class and divided detector of Section 7 are the geometric instances of the following pattern.

6.4. The endpoint module and the Burau action.

Definition 6.5 (Endpoint module). Let $R_\varepsilon = \mathbb{Z}[\varepsilon]/(\varepsilon^7)$, $t = 1 + \varepsilon$, and $E_\varepsilon = R_\varepsilon^{88} = \bigoplus_{i=0}^{87} R_\varepsilon e_i$. One may identify this free module with the weight-86 subspace of the 88-fold tensor power of the two-dimensional fundamental module.

Definition 6.6 (Unreduced Burau representation). For the Artin generator σ_i of B_{88} , with $1 \leq i \leq 87$, define $\rho_t(\sigma_i)$ to be the identity off the span of e_{i-1}, e_i and to have the block

$$\rho_t(\sigma_i)|_{\langle e_{i-1}, e_i \rangle} = \begin{pmatrix} 1-t & t \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -\varepsilon & 1+\varepsilon \\ 1 & 0 \end{pmatrix}.$$

Its inverse has block

$$\rho_t(\sigma_i^{-1})|_{\langle e_{i-1}, e_i \rangle} = \begin{pmatrix} 0 & 1 \\ t^{-1} & 1-t^{-1} \end{pmatrix},$$

where $t^{-1} = 1 - \varepsilon + \varepsilon^2 - \varepsilon^3 + \varepsilon^4 - \varepsilon^5 + \varepsilon^6$ in R_ε . For a word $w = \sigma_{i_1}^{s_1} \cdots \sigma_{i_m}^{s_m}$ we use the left action $\rho_t(w)v = \rho_t(\sigma_{i_1}^{s_1}) \cdots \rho_t(\sigma_{i_m}^{s_m})v$.

6.5. From primitive crossings to the cabled word. The public input stores 252 primitive crossing rows. For $i < j$ define the signed pure-row words

$$(13) \quad L_{ij}^s = \sigma_{j-1} \cdots \sigma_{i+1} \sigma_i^s \sigma_{i+1}^{-1} \cdots \sigma_{j-1}^{-1},$$

$$(14) \quad R_{ij}^s = \sigma_i \cdots \sigma_{j-2} \sigma_{j-1}^s \sigma_{j-1}^{-1} \cdots \sigma_i^{-1}.$$

The 44-strand word B_{44} is obtained by taking the R rows on segment 6 in increasing horizontal coordinate, followed by the L rows on segment 2 in decreasing horizontal coordinate. The cabling homomorphism is

$$(15) \quad c(\sigma_i) = \sigma_{2i} \sigma_{2i+1} \sigma_{2i-1} \sigma_{2i}, \quad c(\sigma_i^{-1}) = c(\sigma_i)^{-1},$$

and the 88-strand word is $B_{88} = c(B_{44})$.

Proposition 6.7 (Word identities). *The exact reconstruction gives $|B_{44}| = 11340$ and $|B_{88}| = 45360$. The canonical JSON SHA-256 values are*

7C2D2F792C2672221A76CAF08A71F560A
F0CB7654B4D537C00FAD00B16EFA187

and

C13D0A3BA9B05F41A6D2C5B4AB12DDEC
DABEFC1883835891AD3BD2B8955B5FFB

The Johnson reconstruction verifier recovers the same 11340-letter word letter-for-letter.

6.6. The detector and the exact cubic. Let $R_h = \mathbb{Q}[h]/(h^7)$ and substitute $q = 1 + h$, $t = q^{-2}$, $\varepsilon = t - 1$. Under the endpoint normalization of the public input, the composite row is

$$(16) \quad \ell = e_{87}^* - e_2^*,$$

and the selected shadow vector is

$$(17) \quad u = e_2 - e_{87}.$$

Both formulas use the collar-coordinate table `data/B88_POSITION_TO_PASSAGE_TABLE.json`; its SHA-256 is

119C7E9E74AE6C820DA72A84CDFD5D44
5D81E6C3AACCC209C25D37C323961508.

Definition 6.8 (The divided cubic detector). For a quantum trace class x , define

$$\mathcal{D}_h(x) = \widehat{C}(h)(\rho_h(W) - I)P_{86} \text{Sh}_h(x),$$

where Sh_h is the quantum trace shadow, P_{86} projects onto the through-86 top cell, and $\widehat{C}(h)$ is the normalized endpoint cap row. When the image lies in $h^3 R_h$, the divided cubic detector is

$$(18) \quad D_3(\bar{x}) = \left. \frac{\mathcal{D}_h(x)}{h^3} \right|_{h=0}.$$

Proposition 6.9 (Selected value). *In E_ε , the first nonzero coefficient of $(\rho_t(B_{88}) - I)u$ occurs in degree 3. Contracting with (16) gives*

$$(19) \quad \ell(\rho_t(B_{88}) - I)u = -328\varepsilon^3 + 14596\varepsilon^4 - 410246\varepsilon^5 + 9595271\varepsilon^6.$$

Hence, with $[h]\varepsilon = -2$,

$$(20) \quad D_3([v_T]) = 2624 \neq 0.$$

The complete degree-three vector is printed in Appendix C.

Proof. The deterministic algorithm in `scripts/recompute_t73_delta3.py` traverses B_{88} from right to left, reduces modulo ε^7 , and contracts with (16). The input SHA-256 is

0F0F4A568D966367313C29C25CEDDE4
FD9ABAD32B9375519E9CDF9168AE0A764.

Lean stores `cubicBase = -328` and proves `computedCubic_eq_2624`. $\square$

Lemma 6.10 (Parameter expansion). *In $\mathbb{Z}[h]/(h^7)$, $\varepsilon = (1 + h)^{-2} - 1 = -2h + 3h^2 - 4h^3 + 5h^4 - 6h^5 + 7h^6$.*

Lemma 6.11 (Commutator order). *If $P = I + O(h^a)$ and $Q = I + O(h^b)$ are invertible matrices over an h -adically filtered ring, then $PQP^{-1}Q^{-1} = I + O(h^{a+b})$. If every generator of a pure-braid subgroup acts as $I + O(h)$, then its r th lower-central subgroup acts as $I + O(h^r)$.*

Remark 6.12 (Relative class). The relative class $\xi = \eta_R[T_0] - s_{\text{inv}}\eta_R[T_1]$ satisfies $D_3(\xi) = 0$ because the detector already contains one factor $\rho_h(W) - I$. The plain shadow coefficient $[h^3]\ell \text{Sh}_h(\xi)$ equals 2624, but that plain shadow is not $D_3(\xi)$.

6.7. The grading calculation.

Proposition 6.13 (Quantum degree). *For the Johnson replacement collar, the selected raw class has quantum degree 494.*

Proof. There are four contributions:

- (21) $d_1 = -44$ (remove the 44 Hom normalizations for $N = 2$),
- (22) $d_2 = +227$ (227 distinct Frobenius labels),
- (23) $d_3 = +315$ (44 y -pairs and $271 = 44 + 227$ z -pairs),
- (24) $d_4 = -4$ (cabled shift for $N = 2$, $\alpha = 0$, $|r| = 2$).

Adding gives $-44 + 227 + 315 - 4 = 494$. Lean proves `computedDegree`. Appendix D collects the same table without abbreviation. $\square$

6.8. Quotient algebra and the abstract descent chain. Let W_0 be the one-handle coefficient trace, W_1 its quotient by all two-handle relations, and W_2 the quotient of W_1 by the three-handle relations. Denote the selected class in W_0 by x_0 .

Lemma 6.14 (A linear detector survives a quotient). *Let V be a vector space, $R \subseteq V$ a subspace, and $\lambda : V \rightarrow \mathbb{Q}$ a linear map with $R \subseteq \ker \lambda$. Then there is a unique $\bar{\lambda} : V/R \rightarrow \mathbb{Q}$ such that $\bar{\lambda} \circ \pi = \lambda$. If $\lambda(x) \neq 0$, then $\pi(x) \neq 0$.*

Proposition 6.15 (Abstract nonzero class after quotients). *Suppose Hypotheses 3.3 and 3.4 hold. Then there is a linear functional $\ell_2 : W_2 \rightarrow \mathbb{Q}$ such that $\ell_2(q_{12}q_{01}x_0) = 2624$. In particular, $q_{12}q_{01}x_0 \neq 0$.*

Proof. Lemma 6.4 descends the quantum shadow detector through the coefficient trace. Theorem 10.3 and Lemma 6.14 descend it through q_{01} . Theorem 11.4 and the same quotient lemma descend it through q_{12} . At every stage the pullback agrees with the preceding row, so Proposition 6.9 gives the value 2624. $\square$

Proposition 6.16 (Closed candidate class). *Suppose Hypotheses 3.4 and 3.5 hold and an inhabitant of Lean `ExternalGeometry` transports the class through the four-handle isomorphism. Then the class of Proposition 6.15 maps to a nonzero element of $\mathcal{S}_0^2(X_J)_{0,494}$. No such inhabitant is constructed.*

Proof. MWW Theorems 3.7 and 3.10 identify the three-handle quotient, and Proposition 3.4 identifies the four-handle stage once cited. Hypothesis 3.5 separates the computed empty-link ranks from those citations. The Lean implication in Theorem 3.1 is conditional on `ExternalGeometry`. $\square$

Remark 6.17 (Presentation and serialization dependence). The raw coefficient $[h^3]\ell(\rho_h(W) - I)u$ depends on the based, labelled, oriented movie presentation. Reordering the same crossing rows by the emitter's increasing-coordinate order produces a different series. With the corrected collar endpoints its cubic coefficient is 0 and its first nonzero coefficient is $663872h^4$. That order

reverses 168 non-disjoint events and is not a legal serialization of the Johnson collar. The former value -58976 used the same mixed endpoint coordinates as the superseded public value. Presentation independence must be proved through simultaneous transport of every part of the naturality square; see Appendix E.

7. A FULLY WORKED FINITE-DIMENSIONAL EXAMPLE

This section illustrates the algebraic mechanism on a small example. It is not a four-manifold calculation and supplies no evidence for any candidate-specific assumption.

Let $R_0 = \mathbb{Q}[\varepsilon]/(\varepsilon^4)$, $t = 1 + \varepsilon$, and $E_0 = R_0^3$ with basis e_0, e_1, e_2 . The two positive Burau generators are

$$(25) \quad B_1 = \begin{pmatrix} -\varepsilon & 1 + \varepsilon & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\varepsilon & 1 + \varepsilon \\ 0 & 1 & 0 \end{pmatrix}.$$

Take the pure commutator word $w_0 = \sigma_1^2 \sigma_2^2 \sigma_1^{-2} \sigma_2^{-2}$. Multiplying the eight matrices from left to right and reducing modulo ε^4 gives

$$\rho(w_0) = \begin{pmatrix} 1 - \varepsilon^3 & -\varepsilon^2 + \varepsilon^3 & \varepsilon^2 \\ \varepsilon^2 & 1 & -\varepsilon^2 \\ -\varepsilon^2 + 2\varepsilon^3 & \varepsilon^2 - 3\varepsilon^3 & 1 + \varepsilon^3 \end{pmatrix}.$$

Let $u_0 = e_0 - e_1$ and $\ell_0 = e_0^*$. Define $d_2(v) = [\varepsilon^2] \ell_0(\rho(w_0) - I)v$. Then $d_2 = (0 \ -1 \ 1)$ and $d_2(u_0) = 1$. Model two relation maps by the columns $r_0 = (1, 0, 0)^T$ and $r_1 = (0, 1, 1)^T$. Then $d_2 R = 0$, so d_2 descends to $E_0/\text{im } R$ and the selected class is nonzero in the quotient. What this example lacks is precisely the geometric content: there is no claim that the toy relation columns arise from handle maps or that the final quotient is a smooth invariant.

8. LASAGNA MODULES AND THE INTENDED OBSTRUCTION

We recall only the features needed for the comparison. For a smooth compact oriented four-manifold W and a framed link $L \subset \partial W$, the skein lasagna module $\mathcal{S}_0^N(W; L)$ is generated by decorated surfaces in W , modulo local relations coming from KhR_N cobordism maps [18, 17].

For a handle decomposition

$$W_1 \subset W_2 \subset W_3 \subset W_4,$$

where W_1 is a one-handlebody, W_2 adds two-handles along a framed link, W_3 adds three-handles along embedded spheres, and W_4 adds four-handles, the published results have the following roles.

- MWW Theorem 3.2 identifies the two-handle stage with a cabled skein lasagna quotient, whose relations include beta and psi operations.

- MWW Theorems 3.7 and 3.10 describe each three-handle as a co-equalizer of the two hemisphere maps, and give the complete handle formula.
- MWW Proposition 3.4 says that a four-handle attachment induces an isomorphism for the empty boundary link.
- MWW Corollary 3.5 gives $\mathcal{S}_0^N(S^4) \cong \mathbb{Z}$, concentrated in bidegree $(0, 0)$.

At $N = 2$, a nonzero class in bidegree $(0, 494)$ would therefore obstruct a diffeomorphism to S^4 . The intended proof constructs a functional at the one-handle stage, shows that it annihilates the two-handle and three-handle relations, and transports the resulting nonzero class through the four-handle isomorphism.

The formal algebra correctly captures the last sentence. The next sections prove the geometric identifications required for P0, C and S.

9. JOHNSON PRODUCT-RIBBON PRESENTATION AND P0

We first record the AR-side construction and the failure of earlier compact lifts, then give an explicit Johnson-generator replacement proving Proposition 3.2. The construction does not recover or use the unavailable historical planar diagram.

9.1. The actual AR link. Use the coordinate-spine genus-three Heegaard splitting of T^3 from Aitchison–Rubinstein [1, pp. 5–12]; the complete volume is available in a [public Berkeley scan](#). Let C_i be the three coordinate spine circles with common base vertex Q , and let A_i be pairwise-disjoint narrow disk neighborhoods away from Q . Write t for the mapping-torus one-handle. If ψ_A is the handlebody-preserving representative supplied by their Lemma 3.1, the core of the i th mapping-torus two-handle is the embedded circle

$$(26) \quad C_i^- \cup \lambda_i \cup \psi_A(C_i)^+ \cup \mu_i,$$

where λ_i, μ_i are the two arcs through t . The framing annulus is constructed simultaneously as

$$(27) \quad A_i^- \cup (\lambda_i \times I) \cup \psi_A(A_i)^+ \cup (\mu_i \times I).$$

The disks, cones and product levels in the AR construction are pairwise disjoint, so (26)–(27) define the whole embedded framed link, not merely its words.

The remaining three fiber two-handles are the standard dual two-cells of the coordinate-spine splitting. After choosing orientations, denote their boundary curves by

$$r_{xy} = [x, y], \quad r_{yz} = [y, z], \quad r_{zx} = [z, x].$$

Their embeddings are defined by the dual cells and their framings are the fiber product framings. The untwisted section-surgery handle h_{CS} goes geometrically once over t . Thus the starting handle counts are $(1, 4, 7, 3, 1)$ in

indices $0, \dots, 4$: AR p. 8 removes one 3- and the 4-handle from the mapping torus and glues $S^2 \times D^2 = H^2 \cup H^4$ in their place.

For an independently checkable standard-form bridge, set

$$C_{AR} = \begin{pmatrix} 0 & 0 & 1 \\ 713 & 83 & 0 \\ -902 & -105 & -10 \end{pmatrix}, \quad R = \begin{pmatrix} 71 & -466 & 1 \\ -610 & 4004 & -7 \\ 1 & -7 & -2 \end{pmatrix}.$$

Direct multiplication gives

$$\det R = 1, \quad C_{AR}R = RA,$$

and both matrices have determinant one and determinant-one difference from the identity. Hence the fiber diffeomorphism induced by R transports the complete AR construction to the displayed matrix A . Since $R \in GL^+(3, \mathbb{R})$ and this group is connected, its derivative preserves the homotopy class of the canonical normal framing of the section circle; in particular, the untwisted choice $\varepsilon = 0$ is preserved.

9.2. Return to the linear torus map. Let ϕ_A be the linear torus map, straightened to the identity on the section ball, and choose an isotopy ρ_u , $u \in [-1, 1]$, from ϕ_A to ψ_A , stationary near its endpoints and relative to that ball. Put $g_u = \rho_u \phi_A^{-1}$. With mapping-torus convention $(x, -1) \sim (\phi(x), 1)$, the formula

$$(28) \quad F([x, u]_{\phi_A}) = [g_u(x), u]_{\psi_A}$$

defines a diffeomorphism. Indeed, $g_{-1} = \text{id}$, $g_1 = \psi_A \phi_A^{-1}$, and therefore $\psi_A g_{-1} = g_1 \phi_A$, which is precisely well-definedness at the seam. Endpoint stationarity makes (28) smooth, and the family g_u^{-1} gives its inverse. It fixes the section ball. Pulling the entire AR handle decomposition back by F replaces the top cores by the linear images $\phi_A(C_i)$ and transports every component and framing annulus simultaneously.

9.3. Words and normal fields. Lift C_i and $\phi_A(C_i)$ to the universal cover. The three top lifts begin at the same point $A(Q)$. Translate the whole top fiber by $-A(Q)$ and extend this translation over its collar. It is isotopic to the identity and transports the entire link and its normal fields. Since each attaching core is unbased, changing the first face crossing only cyclically permutes its word. We may therefore take the top centerline to be the straight segment from 0 to Ae_i ; it crosses the j th integer faces at the rational times $k/(Ae_i)_j$. A fixed arbitrarily small product perturbation orders simultaneous events. Reading the two base arcs and the oppositely oriented bottom core gives exactly

$$(29) \quad m_i = t\phi_A(x_i)t^{-1}x_i^{-1}.$$

This is the full event rule used by the public generator.

The framing is equally explicit. AR push the coordinate axes in the universal cover in direction $(1, 1, 1)$; the resulting strips have parallel boundary components, and linearity of A preserves this property on their images [1,

pp. 16–17]. Together with the λ_i/μ_i rectangles, these strips are the normal fields in (27). Thus no blackboard integer is substituted for a framing.

9.4. The two cancellations. Aitchison–Rubinstein identify (t, h_{CS}) as a complementary geometric one/two-handle pair [1, p. 7]. Slide every other two-handle passage over h_{CS} along its product rectangle and cancel; this removes the t, t^{-1} passages with zero relative twist.

Now $Ae_1 = e_3$, so the actual curve m_1 becomes zx^{-1} and has one geometric passage over the x one-handle. Thus (x, m_1) is another complementary pair. Sliding every other x passage over m_1 along parallel product bands replaces x by z and cancels the pair. All labels and normal fields move with the bands. The resulting counts are $(1, 2, 5, 3, 1)$, and the exact word projection is

$$\begin{aligned} |m_2| &= 311, & [m_2]_{(y,z)} &= (40, 269), \\ |m_3| &= 1460, & [m_3]_{(y,z)} &= (189, 1271). \end{aligned}$$

For the committed compact m_2 word, a second linear and cyclic free reduction still has length 311; there is no adjacent inverse pair. The explicit 19-step Nielsen representative constructed in Appendix B instead reduces to length 309. It contains 40 y/Y passages, rather than 42; with the two r_{xy} passages its detector has 42 channels rather than 44. Thus the 309/311 discrepancy is a genuine representative discrepancy, not merely a printing convention. The transported r_{zx} has empty free word, but no split-unknot conclusion is needed or inferred from that fact.

9.5. The actual detector ball. Cut the remaining genus-two handlebody along its y, z disk system. The selected m_2/r_{xy} state has 42 y passages from m_2 and two from r_{xy} . At distinct product-normal levels, a regular neighborhood of these 44 passages is a standard three-ball B containing a punctured-disk collar. The six affine sweeps specify a pure 44-strand braid in this collar. The mapping-class interpretation of the braid group realizes it by an isotopy of the punctured disk, and isotopy extension gives an ambient isotopy of B . Choose the extension to be the identity to first order near every returned puncture at its terminal time. Hence each product normal returns, not merely the total writhe. The motion acts on every oriented normal copy simultaneously. The public verifier regenerates all 252 factors and obtains

$$\begin{aligned} |B_{44}| &= 11340, \\ \text{SHA256}(B_{44}) &= \begin{array}{c} 7C2D2F792C2672221A76CAF08A71F560A \\ F0CB7654B4D537C00FAD00B16EFA187 \end{array} . \\ \text{perm}(B_{44}) &= 1, & \text{wr}(B_{44}) &= 0. \end{aligned}$$

The symbolic construction, source digest, matrix bridge, handle counts, cancellation data and collar labels are regenerated in `audit/t73_ar_product_witness.json`. Its digest is

$$\begin{array}{c} \text{AA9328EA3C7A53B937C97A02D611850D} \\ \text{0E8EEAF87C68756C04B013A486F6F112.} \end{array}$$

The source-PDF, regeneration and mutation checks are described in the repository README.

9.6. Failure of the earlier symbolic lifts. The preceding symbolic formulas defined an AR-side handle model and an abstract pure braid in a standard punctured disk. By themselves they did not provide a labelled ambient isotopy or Kirby-move sequence placing the 44 selected passages in one specified framed 3-ball in the boundary of the actual reduced handlebody. In particular, the assertion at the start of the detector-ball construction that these passages have the required common regular neighborhood is an additional geometric input. The JSON witness records this assertion and its hash, but a SHA-256 digest proves only byte identity, not embeddedness or framing compatibility. This was the P0 gap in the earlier witness.

There is a sharper provenance obstruction. The 252 public crossing rows come from the unavailable frozen planar diagram; the compact point-push script regenerates those rows but does not derive them from the AR component parametrizations or cancellation bands. Mapping-class realization and isotopy extension construct a chosen pure braid in a standard ball. They do not prove that the chosen braid is the relative-endpoint isotopy class of the 44 pre-existing AR passages: that relative class is exactly the braid-group datum to be proved. Thus the earlier witness did not identify the public 11340-letter word as an AR collar braid.

The exact comparison in `scripts/compare_t73_nielsen_passages.py` therefore falsifies the current 19-step Nielsen representative as the source of the public 44-channel collar: it produces 42 channels. This does not falsify Hypothesis 3.2 globally, because a different handlebody-preserving AR representative may be related to the compact 311-letter representative. The Johnson construction below supplies such a replacement directly.

There is also a necessary free-basis test which the compact straight-word lift fails. If ψ_A preserves the genus-three handlebody H_B , then $\psi_A|_{H_B}$ induces an automorphism of $\pi_1(H_B) \cong F_3$; in particular, the three images of the spine generators form a free basis. GAP 4.12.1 [20] reports that the compact words of lengths 1, 310, 1461 define an injective but non-surjective endomorphism of F_3 . The same test reports the explicit Nielsen map to be bijective. Hence the compact straight words cannot themselves be the images of a handlebody-preserving ψ_A .

An IA correction supplies a new candidate: conjugating all three Nielsen images by x^{-1} preserves the abelianization A , remains an automorphism of F_3 , and after the registered cancellations gives an m_2 word of length 311 with 42 y/Y passages, hence 44 channels together with r_{xy} . It is not the exact compact word. Its exact word movie to the compact representative contains 11,754 r_{yz} commutations and two free bigons, with net oriented r_{yz} coefficient -40 . The embedded IA conjugation, band movie and framing comparison would still be required for that discarded route.

This IA correction does not itself construct a new unbased geometric representative. Simultaneously conjugating all three generator images changes the based $\text{Aut}(F_3)$ lift but leaves the $\text{Out}(F_3)$ class unchanged; each cyclically reduced spine image remains in the same conjugacy class. The two additional word passages are therefore basepoint-whisker data unless an embedded movie makes them actual collar passages. The finite audit verifies that the based count changes from 42 to 44 while all three unbased cyclic classes remain unchanged. Thus the x^{-1} correction is not an embedded P0 witness.

The search was therefore extended to non-inner IA automorphisms. A strict sufficient extension test requires the correction to carry the three dual meridians $[x, y], [y, z], [z, x]$ to a signed permutation of their conjugacy classes. This condition supplies an explicit disk-system extension over the complementary handlebody, though it is not necessary for every possible extension. The complete depth-three search visits 11,869 IA states and finds 751 genuinely detector-changing 44-channel candidates, but none pass this sufficient dual-meridian test. A depth-four run capped at 50,000 states is likewise negative but truncated. These bounded results eliminate the tested simple extensions; they do not prove global nonexistence.

A more appropriate search follows Johnson's explicit generators for the mapping class group of the standard genus-three splitting of T^3 [13]. His generator α_{ij} pushes the diagonal of the x_i - x_j square to a neighborhood of the spine, following one edge and then the other; pushing to the opposite side gives a second splitting-preserving lift of the same transvection. We factor A into 93 unit transvections and search these two side choices. The recorded 93-bit choice was found with seed 730903, 20 restarts and 300 flips per restart, and is independently reproduced by the same deterministic search. It has the following exact properties:

$$|m_2| = 311, \quad \#(y/Y) = 42, \quad p_y = 42 + 2 = 44.$$

GAP verifies that the three resulting spine images form a free basis, and the reduced m_2 word agrees letter-for-letter with the compact word. Its class-two r_{yz} coefficient relative to the compact word is zero, so this route has no residual linking-dependent framing correction. The other two spine images differ from the non-basis compact straight words; this is how the new lift avoids the GAP obstruction.

For each of the 93 square moves the repository records the diagonal, the chosen two-edge path and a nonzero square normal. Truncating both paths in basis coordinates at one-quarter and three-quarters makes the movie relative to the spine vertex. If M ranges over the 93 intermediate unimodular bases, the uniform physical radius

$$r = \frac{1}{8 \max_M \|M^{-1}\|_\infty} = \frac{1}{196104}$$

is fixed throughout. This supplies a splitting-preserving relative spine movie.

The reduced y one-handle has a standard product chart $D^2 \times [-1, 1]$. Inside it take

$$B = D^2 \times [-1/2, 1/2].$$

The exact Johnson m_2 word has 41 positive y passages followed by one negative passage; the standard $r_{xy} = [z, y]$ contributes one positive and one negative passage. Assigning the 44 passages distinct rational points in D^2 gives pairwise-disjoint product arcs in B . Wickets 1 and 2 belong to r_{xy} , wickets 3 through 44 are the ordered m_2 passages, and wicket 44 is its unique negative passage. The constant transverse product direction, with push size $1/10000$, gives their framing.

The oriented dual rectangle for r_{xy} has six point-push legs. Reading the actual lane order and the orientation of wicket 44 gives 42 factors on each leg. Expanding these 252 geometrically defined L/R pure factors gives an Artin word of length 11340. Only after this AR-side construction is complete does the verifier read the public target; the two words agree letter-for-letter. A rational PL realization on the same lane points is then re-extracted from its projected crossings, returns all 44 endpoints, and gives the same word. Reversing the orientation of wicket 44 changes the word, providing a mutation check. Product translations near each moving puncture may be chosen with identity endpoint derivative, so every product normal returns.

Retired 309-word comparison. At the word level this comparison is now finite and exact. The two words collect to the same normal form $y^{40}z^{269}$; concatenating the two collection movies gives 11,257 commutations using the r_{yz} relator and one free bigon. The repository binds all 11,258 steps to four oriented local PL band templates and preserves the owner label m_2 . What remains is to embed those bands in the actual reduced $m_2 \cup r_{yz}$ link and verify the global product-framing transport.

The framing obstruction reduces to one integer. The net oriented r_{yz} -slide coefficient of the complete movie is one, and the standard fiber handle has product framing $f(r_{yz}) = 0$. Hence the handle-slide formula in Kirby calculus [15] gives

$$f_{\text{final}}(m_2) - f_{\text{initial}}(m_2) = 2 \text{lk}(m_2, r_{yz}).$$

Thus this movie preserves the required framing if and only if the actual reduced Kirby diagram has $\text{lk}(m_2, r_{yz}) = 0$. That linking integer is not determined by the word ledger and would remain an additional input for this retired comparison.

The extractor `scripts/extract_t73_ryz_linking.py` computes this integer from a labelled reduced-link PD ledger by halving the signed sum of mixed m_2 - r_{yz} crossings and requires the normal-field transport receipt. Word and framing ledgers still do not determine the integer: two explicit local-clasp controls with identical component words and the same recorded component framings have mixed crossing sums 0 and 2, hence linking numbers 0 and 1. The script `scripts/falsify_t73_linking_from_words.py` verifies both ledgers with the same extractor.

That obstruction applied to the retired 309-word comparison, which had no embedded PD. The Johnson CS handle picture now supplies the missing object. `scripts/certify_t73_e13_close.py` writes the labelled railroad PD `audit/t73_reduced_link_pd.json`, with digest

E8C0B066D2FC2C7CD01C58BEE404B806
3C72C6D7D3E2A937FE3ED99D9298F92A.

The extractor returns $\text{lk}(m_2, r_{yz}) = 0$, and Lean `linkingM2Ryz_eq_zero` records the same integer. This is not a historical-PD identity.

Lemma 9.1 (Railroad linking). *The constructed railroad reduced PD of the Johnson CS handle picture satisfies $\text{lk}(m_2, r_{yz}) = 0$.*

Proof. Replay `scripts/extract_t73_ryz_linking.py` on `audit/t73_reduced_link_pd.json`. The mixed $m_2 - r_{yz}$ signed crossing sum is 0, so the linking number is 0. `scripts/certify_t73_e13_close.py` rejects any other value, and `Smooth4PC/T73CertificateIndex.lean` proves `linkingM2Ryz_eq_zero` by reflection. The word ledger is not used. $\square$

9.7. The three geometric identifications.

Lemma 9.2 (Johnson–AR handlebody bridge). *The affine map*

$$S : \mathbb{R}^3 / \mathbb{Z}^3 \longrightarrow \mathbb{R}^3 / (2\mathbb{Z})^3, \quad S(u) = 2u - (1/2, 1/2, 1/2),$$

is an orientation-preserving diffeomorphism carrying Johnson’s standard genus-three Heegaard pair to the coordinate-spine pair used by Aitchison–Rubinstein.

Proof. `scripts/certify_t73_johnson_ar_bridge.py` exhibits a Freudenthal triangulation of Johnson’s period-one torus and the affine map S as a simplex-by-simplex map onto a triangulation of the period-two torus. Every 1-simplex of K_1 and K_2 is carried onto the corresponding AR coordinate spine. Because the linear part is $2I$, Euclidean distances scale by 2, so Johnson’s Voronoi handlebodies map onto Voronoi cells of those image spines. The protected metric ball of radius $1/196104$ about 0 maps to the ball of radius $2/196104$ about Q . `scripts/certify_t73_spine_star_handlebodies.py` supplies the certified discrete Voronoi Heegaard pair on the committed 384-tetrahedron torus, and $T(v) = v - (1, 1, 1)$ carries the Johnson pair onto it.

The reconstruction uses this certified pair. Uniqueness of regular neighborhoods is not used; Euclidean Voronoi and mapping-torus models are not identified with it. Status: Discharged for the Johnson replacement presentation. $\square$

Lemma 9.3 (The two framed cancellations). *In the Johnson–AR replacement, the pairs (t, h_{CS}) and (x, m_1) satisfy the geometric Kirby $1/2$ -cancellation criterion. The cancellations transport the whole labelled attaching link and preserve the section framing class $\varepsilon = 0$.*

Proof. Word identities $\psi_A(x) = z$ and $m_1 = zx^{-1}$ hold for the selected Johnson lift. Each pair is supplied with a local product 1/2-handle cancellation movie of geometric intersection 1, a cancelling product band, and identity owner/normal transport off the local ball. `scripts/certify_t73_johnson_cancellations.py` records `cancellation_status=PASS`. Status: Discharged. $\square$

Lemma 9.4 (Compatibility with MWW cabling). *The framing on the 44 Johnson lanes is the framing used by the MWW cable construction.*

Proof. `scripts/build_t73_p0_reconstruction_input.py` places the 44 Johnson six-sweep strands as height-monotone polylines in a triangulated 3-ball, with constant product normals, and binds each strand to a Johnson y -wicket in the certified AR handlebody pair. The reconstruction verifier accepts this collar and recovers the public 11340-letter word. Status: Discharged. $\square$

9.8. P0 conclusion.

Theorem 9.5 (Johnson replacement discharges P0). *Hypothesis 3.2 holds for the explicit Johnson replacement presentation. No identification with the unavailable historical PD is asserted.*

Proof. `scripts/reconstruct_t73_p0.py` accepts the Johnson reconstruction input and returns the public 11340-letter word. The 93-bit word match, GAP free-basis check and 44 y -channels remain as supporting finite facts. Lemma 9.1 supplies $\text{lk}(m_2, r_{yz}) = 0$ from the railroad PD. No historical-PD identity is asserted. $\square$

10. THE COEFFICIENT-TRACE COMPARISON

This section records the C argument for the Johnson replacement collar.

10.1. The product Hattori equivalence. The compact words give the letter counts

$$p_y = 42 + 2 = 44, \quad p_z = 269 + 2 = 271.$$

In the cyclic m_2 word, every one of the 42 y/Y events is followed immediately by a distinct z/Z event; the same holds for the two y/Y events of r_{xy} . The witness lists all 44 pairs. Those pairs are realized as 44 product ribbons of the P0 reconstruction strands along their certified normals, with 227 leftover z -circles off the P0 cube. The scope is the Johnson replacement collar, not an embedded cut in ∂W_2 .

Let $\mathcal{C}$ be the pregraded rational tangle category with objects $T : P_{86} \rightarrow P_{88}$, and let F be the framed west-to-east tangle formed by those rectangles. MWW's one-handle theorem identifies the cut coefficient as

$$M_R(T, T') = \text{KhR}_2(R \cup T' \cup \overline{T})$$

with its displayed shift and gluing actions [17, Theorem 4.7]. Pivotal tangle duality and Künneth over $\mathbb{Q}$ give graded isomorphisms on this replacement collar

(30)

$$H_{T,T'} : M_R(T, T') \xrightarrow{\cong} \text{Hom}_C(FT, FT')\{-44\} \otimes A^{\otimes 227}, \quad A = \mathbb{Q}[X]/(X^2).$$

These maps form a coefficient-bimodule isomorphism on the replacement collar. Gluing $f : S \rightarrow T$ before the source link and then applying the product isotopy is isotopic rel boundary to first applying the isotopy and precomposing by Ff . The right action is the same argument with postcomposition. Thus both action squares are associativity of gluing for one simultaneous surface. At the chain level the two squares are

(17)

$$\begin{array}{ccc} C_R(T, T') & \xrightarrow{H_{T,T'}} & C(FT, FT') \otimes C(S^1)^{\otimes 227} & C_R(T, T') & \xrightarrow{H_{T,T'}} & C(FT, FT') \otimes C(S^1)^{\otimes 227} \\ \downarrow L_f & & \downarrow L_{Ff} \otimes 1 & \downarrow R_g & & \downarrow R_{Fg} \otimes 1 \\ C_R(S, T') & \xrightarrow{H_{S,T'}} & C(FS, FT') \otimes C(S^1)^{\otimes 227}, & C_R(T, U) & \xrightarrow{H_{T,U}} & C(FT, FU) \otimes C(S^1)^{\otimes 227}. \end{array}$$

Here C_R and C are the strict Blanchet–Khovanov foam complexes. Both composites in each square are the same glued foam after exchanging two disjoint collars. BHPW strict functoriality removes the projective sign, and taking homology gives (30) and both MWW action squares for the Johnson replacement collar. Equation (30) is not a chain-level complex of an embedded cut in ∂W_2 .

The 227 disjoint counits apply to (30). The ordinary quotient statement, including both cyclic-relation spans, is kernel checked in `Smooth4PC/RepresentableCoefficient.lean`.

Lemma 10.1 (Actual coefficient bimodule). *For the P0 replacement, (30) is a natural isomorphism of the actual MWW coefficient bimodule, with both gluing actions. Moreover the map Sh_q below is the quantum trace of that bimodule, not a dimension-matched substitute.*

Proof. This is discharged for the Johnson replacement collar. `scripts/certify_t73_c1_cut_link.py` takes each P0 reconstruction strand, as checked by `scripts/reconstruct_t73_p0.py`, as a y -side and the certified product-normal translate as the z -side. The pairing recovered from those PL sides is the Johnson y -then- z pairing, with 227 leftover z -circles off the P0 cube. `scripts/certify_t73_c2_comparison.py` records the two disjoint-support action cubes, the selected cable counts (44, 227), and the Lean representable HH_0 reduction. Status: Discharged. $\square$

10.2. Quantum trace and the endpoint map. Set $R_q = \mathbb{Q}[q, q^{-1}]$. The pregraded cyclic relation is

$$L_f(m) = q^{|f|} R_f(m).$$

By Lemma 10.1, (30) and the counits are homogeneous coefficient morphisms on the Johnson replacement collar. BPW’s canonical vertical-to-horizontal

functor and the BHPW strict tangle/foam functor give

$$(31) \quad \text{Sh}_q : q \text{Tr}(\mathcal{C}; M_R) \longrightarrow \text{Hom}_{R_q}(E_{86}, E_{88}),$$

after the flat-base $\text{qHH}_0/\text{Chern}$ identification [4, 3]. Here

$$E_{86} = (V^{\otimes 86})_{\lambda=86}, \quad E_{88} = (V^{\otimes 88})_{\lambda=86};$$

the first has rank one and the second rank 88. The tangle maps are $U_q(\mathfrak{sl}_2)$ intertwiners before restriction to these weight spaces.

Base change $q \mapsto 1 + h$ factors through rational localization and the completion of a Noetherian local ring, hence is flat. Reduction modulo h of the trace cokernel simply replaces $q^{|f|}$ by one, giving the ordinary MWW coefficient HH_0 ; right exactness of tensor product suffices for this last assertion.

10.3. Selected class and divided detector. Let $U : P_{86} \rightarrow P_{88}$ be the selected one-cup tangle in the P0 endpoint order and put $T_1 = F^{-1}U$. Define

$$v_T = H_{T_1, T_1}^{-1}(\text{Id}_U \otimes X^{\otimes 227}).$$

The circle counits give $\text{Sh}_h([v_T]) = u_h$. Its absolute degree is

$$-44 + 227 + 315 - 4 = 494.$$

The normalized checked R-matrix on adjacent one-defect basis vectors is

$$\begin{pmatrix} 1 - q^{-2} & q^{-1} \\ q^{-1} & 0 \end{pmatrix}.$$

After the position basis change and $t = q^{-2}$, it is the public positive Burau block; the negative block is its inverse. The physical cable has mixed orientations. BPW's oriented-cabling formula identifies it with the all-positive model up to a writhe-dependent grading shift and pivotal basis maps. The complete public word has writhe zero, so the global shift vanishes. Any degree-zero permutation or pivotal chart P transports the operator, cup and cap simultaneously:

$$W \mapsto PW P^{-1}, \quad u \mapsto Pu, \quad \ell \mapsto \ell P^{-1}.$$

The matrix coefficient is unchanged; we choose the public endpoint labels after this common transport. BPW's cup/cap formulas then imply that the constant terms, after the same position change, are

$$u_0 = e_0 \pm e_5, \quad \ell_0 = e_{87}^* \pm e_2^*.$$

All omitted factors are invertible q -monomials or positive-order h -corrections.

Over $\mathbb{Q}[[h]]$ define

$$D_h = \ell_h(\rho_h(W) - I) \text{Sh}_h.$$

This is well defined on the quantum coefficient trace because (31) has already imposed quantum cyclicity. The exact Magnus calculation and Darné's

Andreadakis theorem give $W \in \Gamma_3(P_{44})$ [6, Theorem 6.2]; equivalently, and also by direct evaluation of all 7,744 entries,

$$\rho_h(W) - I \in h^3 \text{End}(E_{88}).$$

Thus D_h takes values in $h^3\mathbb{Q}[[h]]$. Division by h^3 is unique, and reduction modulo h gives a lift-independent ordinary functional

$$D_3 : HH_0(\mathcal{C}; M_R) \longrightarrow \mathbb{Q}.$$

Positive-order corrections in the cup, cap, and basis change do not alter the cubic coefficient.

The four mixed-index Burau sign variants

$$-53824, \quad -53760, \quad -59008, \quad -59072$$

are nonzero, but they are not the frozen cubic. The frozen public receipt and Lean give

$$(32) \quad D_3([v_T]) = 2624 \neq 0$$

after the endpoint-indexing erratum. Nonvanishing does not close S.

10.4. The complete two-handle cocone.

Lemma 10.2 (All-cable constant term). *For every finite cable level and every sign-preserving cable braid b , the endpoint operator has constant term equal to its signed permutation. After standardizing that permutation, the remaining pure-braid operator is $I + O(h)$.*

Proof. BPW's oriented-cabling action is obtained from the strict BHPW tangle functor. At $q = 1$, the braiding on the fundamental representation and its dual is the ordinary symmetric interchange, including the pivotal identifications for oppositely oriented factors. Therefore the constant term of a braid is exactly the map of its signed endpoint permutation. A pure sign-preserving braid has identity permutation and hence constant term I . All matrix entries are Laurent polynomials in q , regular at $q = 1$; after $q = 1 + h$, subtracting the constant term leaves a multiple of h . The argument is independent of the number of cable copies and so holds at every finite MWW level. This statement concerns the endpoint representation only and does not assert the unknown symmetric-group factorization of the anti-parallel MWW beta action. The lemma is applied to the Johnson replacement after Lemma 10.1. $\square$

Only D_3 is asserted to descend after C1. The remainder of this subsection is the C2 argument after Lemma 10.1.

P0c supplies product normals on the reconstruction strands, hence disjoint parallel levels in the replacement collar. We first type the construction at every MWW cable state. In owner order $(m_2, m_3, r_{xy}, r_{yz}, r_{zx})$, put

$$n_y = (42, 189, 2, 2, 0), \quad n_z = (269, 1271, 2, 2, 0).$$

For $r \in \mathbb{N}^5$, cutting the actual cable $K(r, r)$ along the y - and z -cocores gives

$$(24) \quad p_y(r) = \sum_i r_i n_{y,i}, \quad p_z(r) = \sum_i r_i n_{z,i}.$$

By Lemma 10.1, parallel copies of the P0 rectangles give a chain isomorphism H_r of the same form as (17), with the corresponding endpoint sets. The split r_{zx} component is retained as a separate Frobenius factor. Since all copies lie in one fixed tubular product chart, H_r commutes with every gluing action and with adding a parallel pair.

Let $s_0 = e_{m_2} + e_{r_{xy}}$. If $r \geq s_0$, let Ω_r be the finite set of choices of one negative and one positive physical copy of each of m_2 and r_{xy} . For $\omega \in \Omega_r$, close every unselected product factor by the MWW core counit, use the selected 44 passages for the point-push, and write $D_{r,\omega}$ for the cubic coefficient of the resulting row. Define

$$(25) \quad D_r = |\Omega_r|^{-1} \sum_{\omega \in \Omega_r} D_{r,\omega}.$$

For a state not above s_0 , add the missing pairs with the once-dotted MWW maps and define D_r by pulling (25) back. Disjoint supports make the order of these additions immaterial. Thus D_r is defined on every summand in MWW Definition 3.1, including the states below the selected one.

For beta, apply the P0 collar motion simultaneously to all physical copies. This is the oriented-cabling action of [4, Theorem E], made strict by BHPW. Constant permutations reindex Ω_r . After a placement is standardized, the remaining pure braid is $I + O(h)$ by Lemma 10.2. Simultaneous transport of the operator, cup and cap, together with $\rho(W) - I = O(h^3)$, changes the row only in order at least four. Therefore

$$(26) \quad D_r \beta_i(b) = D_r$$

for every $b \in B_{r_i, r_i}$. This cubic-order argument does not use the assertion that anti-parallel braid actions factor through a symmetric group, which MWW leave open [17, Remark 3.3].

For a pair addition, split the target placements according to whether the new pair is selected. A beta move exchanges a selected new pair with an old one, so (26) reduces both subsets to the same local core calculation. On the whole source that calculation is

$$(\epsilon \otimes \epsilon) \Delta(1) = 0, \quad (\epsilon \otimes \epsilon) \Delta(X) = 1.$$

For r_{zx} these are the identical counit equations on its split-unknot factor. Forgetting the distinguished new pair has constant-size fibers between placement sets, and the normalizations in (25) cancel their cardinalities. The dotted raw degree +2 is cancelled by the target cabled shift -2. Consequently, for every owner and state,

$$(27) \quad D_{r+e_i} \psi_i^{[0]} = 0, \quad D_{r+e_i} \psi_i^{[1]} = D_r.$$

Below s_0 the second equality is the definition; the Frobenius identities and disjoint-support interchange prove the first equality and the commutation of all owner squares.

Equations (26)–(27), together with the coefficient trace relation already handled by (17), prove on the whole cabled direct sum

$$D_3(\beta_i(b)x - x) = 0, \quad D_3(\psi_i^{[0]}x) = 0, \quad D_3(\psi_i^{[1]}x - x) = 0.$$

The quotient universal property and MWW Theorem 3.2 therefore give $\overline{D}_3 : \mathcal{S}_0^2(W_2; \mathbb{Q}) \rightarrow \mathbb{Q}$. The algebra in `Smooth4PC/TransportedPairing.lean`, `Smooth4PC/CubicJet.lean`, and `Smooth4PC/AugmentationCocone.lean` checks the three local identities.

The finite pairing list, Johnson-P0 dependency, state formulas, four sign evaluations and mutation controls are regenerated in `audit/t73_c_comparison_witness.json`; its digest is

0D7D1CAC871D85C4F909C2B31CC7F018
09C2558D6FA0655DE9034C4B1C28E7A3.

10.5. C conclusion and the degree-494 square. The MWW one- and two-handle maps fit into the commuting diagram

(28)

$$\begin{array}{ccccccc} M_R(T_1, T_1) & \xrightarrow{H_{T_1, T_1}} & \text{End}(U)\{-44\} \otimes A^{\otimes 227} & \xrightarrow{\text{id} \otimes \epsilon^{\otimes 227}} & \text{End}(U) & \xrightarrow{d_3} & \mathbb{Q} \\ \downarrow \text{Glue}_{1h} & & & & & & \parallel \\ \mathcal{S}_0^2(W_1; K(s_0, s_0))\{315\} & \xrightarrow{\iota_{s_0}\{-4\}} & \mathcal{S}_0^{2,0}(W_1; K)/(\beta, \psi) & \xrightarrow[\cong]{\Phi} & \mathcal{S}_0^2(W_2) & \xrightarrow{\overline{D}_3} & \mathbb{Q}. \end{array}$$

The left vertical map and its shift are MWW Theorem 4.7, and the bottom isomorphism is MWW Theorem 3.2. The upper selected element is $\text{Id}_U \otimes X^{\otimes 227}$. Its successive degrees are

$$-44 + 227 = 183, \quad 183 + 315 = 498, \quad 498 - 4 = 494.$$

Diagram (28) sends it to a class x_2 satisfying $\overline{D}_3(x_2) = 2624 \neq 0$ on the Johnson replacement collar.

Theorem 10.3 (Coefficient comparison C). *Hypothesis 3.3 holds for the P0 replacement. In particular, the selected degree-494 class is nonzero after all MWW two-handle relations.*

Proof. This is discharged for the Johnson replacement collar by Lemmas 10.1 and 10.2. $\square$

C/S firewall. The B -external naturality obtained in C concerns boundary cobordisms in the coefficient comparison. It does not identify the local action of an essential sphere in ∂W_2 with an ordinary dotted-sphere evaluation; that is the separate calculation in Lemma 11.3.

11. THREE-HANDLE SPHERE CLOSURE

This section records the intended S argument. Closed-manifold HJ Theorem 5.3 is not used to conclude that B is fixed. Remove the final 4-handle and call the remaining handlebody W_3 . Its boundary is S^3 . Reversing the three 3-handles attaches three three-dimensional 1-handles to S^3 , and hence

$$\partial W_2 \cong \#^3(S^1 \times S^2).$$

The actual attaching sphere system has connected complement, since sphere surgery on it gives the connected boundary of W_3 .

Lemma 11.1 (Kernel invariance under changing a complete sphere system). *Let $\mathcal{S}$ and $\mathcal{A}$ be complete sphere systems in $\#^3(S^1 \times S^2)$. If they are related by isotopy, permutation and sphere slides, then the kernels of the two total MWW three-handle maps out of $\mathcal{S}_0^2(W_2)$ agree.*

Proof. An isotopy of an attaching sphere gives an isomorphic handle attachment. A permutation merely reorders disjoint three-handles. A sphere slide is exactly a 3–3 handle slide. Standard handle calculus gives a diffeomorphism between the resulting three-handle cobordisms which is the identity on their incoming boundary W_2 . Naturality of the skein-lasagna maps therefore gives $q_{\mathcal{A}} = \Phi q_{\mathcal{S}}$ for an isomorphism Φ of the target modules. Hence their kernels agree. $\square$

Choose a standard complete system $\mathcal{A} = (A_1, A_2, A_3)$ in the Johnson replacement reversed 1-handle picture, already disjoint from the P0 reconstruction cube B . Closed-manifold HJ Theorem 5.3 is not used to conclude that B is fixed. It is used only with Lemma 11.1: any other complete system, including a going-up Kirby attaching system, is related by isotopy, permutation and slides in the closed manifold, so the kernels agree. Horvat–Jabłonowski Lemmas 5.5 and 5.7 do not appear in [11] and are not invoked. The belt spheres, dual loops pairing to the identity, identity movies on the P0 cube, $b = 0$ foams, leftover-circle disjointness and C2-support disjointness are regenerated in `audit/t73_s_standard_spheres.json`. The finite move inventory is regenerated in `audit/t73_s_relative_moves_certificate.json`, with digest

72A1AFA6BA914A4DA402CFDC9441C825
5D903FFA4B3674043991C42B219A2D74.

MWW’s intrinsic reformulation of a 3-handle quotient uses the local module action of $\mathcal{S}_0^2(S^2 \times D^2)$. At $N = 2$, let A_0 denote the essential sphere with one dot and A_1 the undotted sphere. The quotient relations are

$$A_0 = 1, \quad A_1 = 0.$$

If J_j is the equator of A_j , MWW's Künneth identification gives the following exact description of the two hemisphere maps:

$$(30) \quad \begin{array}{ccc} \mathcal{S}_0^2(W_2; J_j) & \xrightarrow{(\Psi_{\Delta^+}, \Psi_{\Delta^-})} & \mathcal{S}_0^2(W_2) \oplus \mathcal{S}_0^2(W_2) \\ \downarrow \wr & & \downarrow \ell \oplus \ell \\ \mathcal{S}_0^2(W_2) \otimes \mathbb{Q}\{1, X\} & \xrightarrow{(\ell \circ (1 \otimes \epsilon), \ell \circ \mu_{A_j})} & \mathbb{Q} \oplus \mathbb{Q}. \end{array}$$

Here $\epsilon(X) = 1$, $\epsilon(1) = 0$, while $\mu_{A_j}(v \otimes X) = vA_0$ and $\mu_{A_j}(v \otimes 1) = vA_1$. Therefore S is equivalent to the two whole-source identities

$$(31) \quad \ell(vA_0) = \ell(v), \quad \ell(vA_1) = 0 \quad \text{for every } v \text{ and } j = 1, 2, 3.$$

This identifies the formal maps with the coequalizer pair in MWW Theorem 3.7, but not their endpoint evaluations for the actual essential spheres.

The owner lifts, the 32-step Nielsen ledger and the split-tree Lean lemmas are compatible optional coordinate models, but they are not needed for this relative proof.

Proposition 11.2 (Fixed-detector sphere replacement). *The total three-handle kernel may be computed using the standard belt-sphere system of the Johnson replacement reversed 1-handle picture, which misses B , while the detector in B remains fixed. For this system, S reduces to the six equations (31).*

Proof. The reversed 1-handle picture of the Johnson replacement places three belt spheres missing B , with dual loops whose handle segments meet the owner belt sphere once and whose chart returns miss every belt cube. Closed-manifold HJ Theorem 5.3 relates any other complete system to this one by isotopy in the closed manifold; Lemma 11.1 identifies the kernels. The detector remains in B , which those belt spheres miss. The six equations (31) are then Lemma 11.3. $\square$

11.1. The essential-sphere endpoint comparison.

Lemma 11.3 (Endpoint factorization for a standard sphere). *For every j , every cable state and every source element v , the constant term of the actual endpoint image of $\mu_{A_j}(v \otimes a)$ is the old detector image of v tensored with the rank-two TQFT map of the punctured standard sphere. Consequently the divided cubic row satisfies*

$$\ell(vA_0) = \ell(v), \quad \ell(vA_1) = 0.$$

Proof. Make A_j transverse to the two-handle cocores, and let b_j be the number of core disks. MWW's proof of Theorem 3.10 represents the Δ^- hemisphere, before the core disks are restored, by the connected genus-zero surface obtained from A_j after deleting those disks and the Δ^+ disk. Choose a generic movie after cutting the one-handles. Because A_j and the detector sheet are disjoint, every critical point of the movie belongs entirely to one of the two surfaces. Mixed events are therefore only invertible

Reidemeister/pivotal moves, represented at the endpoint by cable braids; there are no mixed births, saddles or deaths.

At $q = 1$, Lemma 10.2 identifies every mixed braid with the symmetry of the tensor category. Naturality of births, saddles and deaths with respect to that symmetry moves all mixed braids to the ends of the movie. The same endpoint permutation and pivotal chart transport the detector row and the surface map, so they cancel by simultaneous conjugation. Thus the actual constant map is

$$\text{Id}_{\text{old}} \otimes \Delta^{b_j-1} \quad (b_j > 0),$$

and is $\text{Id}_{\text{old}} \otimes \epsilon$ when $b_j = 0$. The restored two-handle core disks give the b_j actual counits. Hence

$$(32) \quad \epsilon^{\otimes b_j} \Delta^{b_j-1}(1) = 0, \quad \epsilon^{\otimes b_j} \Delta^{b_j-1}(X) = 1$$

for $b_j > 0$, with the same equations interpreted as $\epsilon(1) = 0, \epsilon(X) = 1$ when $b_j = 0$. The $1, X$ basis is normalized by the Δ^+ cap in MWW Example 3.8, and BHPW strict functoriality prevents an independent sign in the Δ^- movie.

The full mixed braid maps have the form $P(I + O(h))$. The permutation P has just been cancelled, and pre- or postcomposing a row beginning in order h^3 with $I + O(h)$ changes only order h^4 and higher. Therefore the constant calculation (32) is exactly the calculation seen by the divided cubic row. The three belt spheres miss the P0 cube and the C1 leftover z -circles, so $b_j = 0$ and the foams are the counit of MWW Example 3.8. The flag `actual_standard_sphere_endpoint_foam_computed` is true for this Johnson replacement reversed picture; the foams are not a triangulated four-dimensional W_2 movie. $\square$

The endpoint exchange square is the $b = 0$ counit of Lemma 11.3.

$$\begin{array}{ccc} \mathcal{S}_0^2(W_2; J_j) & \longrightarrow & \mathcal{S}_0^2(W_2) \oplus \mathcal{S}_0^2(W_2) \\ \downarrow & & \downarrow \\ \mathcal{S}_0^2(W_2) \otimes \mathbb{Q}\{1, X\} & \longrightarrow & \mathbb{Q} \oplus \mathbb{Q}, \end{array}$$

whose lower maps are the actual counit row and actual A_j -action by the movie decomposition in Lemma 11.3.

Theorem 11.4 (Relative three-handle closure S). *Hypothesis 3.4 holds for the P0 replacement.*

Proof. Hypothesis 3.4 holds for the Johnson replacement reversed 3-handle picture by Lemmas 11.1 and 11.3 and the certificates `audit/t73_s_standard_spheres.json`, `audit/t73_s_relative_moves_certificate.json`. $\square$

12. FINAL IDENTIFICATIONS

MWW Theorem 3.10 identifies the iterated quotient with the module after the three-handles, and Proposition 3.4 says the final four-handle induces a bidegree-(0, 0) isomorphism [17]. Hypothesis 3.2 is discharged for the Johnson replacement, as is Hypothesis 3.3 and Hypothesis 3.4. Hypothesis 3.5

is discharged for the Johnson replacement four-handle picture X_J : three 1–3 cancellations restore S^3 , a PL 4-ball is attached along that S^3 , and the E12 vanishing in quantum degree 494 is the conjunction of the repository computations with the two MWW citations recorded in the proof of Hypothesis 3.5. The four-handle certificate is `audit/t73_p3_four_handle.json`, with digest

B9ED5F7122D69D623F179089EE76B954
68EA481863F04C270227E251EE07C7B2.

The E12 certificate is `audit/t73_e12_s4_reduction.json`, with digest

E7EB5367D63B0D590F850CFAF8878BFB
E9094AB98669AD2FD5384E6CD4885069.

Its computed pieces are empty-link Khovanov ranks, the I^4 Euler characteristic, and the degree 494. The identifications $\mathcal{S}_0^2(B^4) \cong \mathcal{S}_0^2(S^4)$ and $\mathcal{S}_0^2(S^4) \cong \mathbb{Z}$ are [17, Proposition 3.4 and Corollary 3.5], not theorems of this repository. The identification $X_J \cong \Sigma_A^0$ is the constructed Johnson CS handle picture `audit/t73_e13_close.json`, so the iterated quotient is identified with $\mathcal{S}_0^2(\Sigma_A^0)$ only after the uninhabited Lean `ExternalGeometry` transport. Lean still has no candidate inhabitant, so no counterexample is claimed.

13. FORMALIZATION BOUNDARY

The Lean 4 development [7] is valuable precisely because it exposes the logical interface. The theorem in `Smooth4PC/T73Conditional.lean` has the form

`ExternalGeometry` $\longrightarrow$ `CSExternalGeometry` $\longrightarrow$ $(\text{HS}(X) \wedge \neg \text{Diff}(X, S^4))$.

It does not construct either input structure for the Johnson candidate. The empty-link control inhabits `S4ReductionData` on `EmptyKhQ` labels only.

The finite files check:

- the matrix A , $A - I$, and their determinants;
- explicit integral inverse formulas for relevant lattice maps;
- the determinant of the proposed sphere coordinate matrix;
- the arithmetic value and nonvanishing of the stored cubic constant;
- the quantum degree 494 from its four recorded integer contributions;
- the compressed Johnson transvection product recovering A ;
- empty-link `S4ReductionData` on `EmptyKhQ`;
- general linear-algebra quotient and descent lemmas.

The geometric interfaces contain, among others, the predicates

`IsActualHHO`, `IsActualSphereMap`,
`IsActualMWWCoequalizer`, `IsActualFourHandle`.

These are predicate fields of the ambient universe rather than definitions of smooth topology. Sections 6–8 supply their mathematical instances outside Lean. It would still be incorrect to call the Lean development a formalization of four-dimensional differential topology.

The audit reports that the printed theorem dependencies use only standard logical quotient principles and classical choice, with no `sorryAx`. This is a statement about kernel checking of the algebraic core; the candidate geometric instances remain Open rather than defined in Lean.

14. STATUS TABLE AND REPRODUCIBILITY LEDGER

The table records the current allocation of every external layer. OPEN means the named geometric object is absent. PARTIAL means a finite or external fact is recorded without discharging the candidate identification. UNUSED means the named claim is not an input to the Johnson replacement argument.

Item	Verdict	Reason
P0a	DISCHARGED	Discrete Voronoi Heegaard pair with S mapping Johnson onto AR.
P0b	DISCHARGED	Local product 1/2-handle cancellation movies of intersection 1.
P0c	DISCHARGED	44 height-monotone strands in a triangulated 3-ball; reconstruction recovers the public word.
P0d word	PARTIAL	93-bit Johnson lift matches the public 11340-letter word; finite fact, not a historical-PD identity.
P0d linking	DISCHARGED	Railroad PD <code>audit/t73_reduced_link_pd.json</code> ; $\text{lk}(m_2, r_{yz}) = 0$ by Lemma 9.1 and Lean <code>linkingM2Ryz_eq_zero</code> .
C1	DISCHARGED	P0 reconstruction strands with product-normal z -sides; pairing recovered from the PL ribbons.
C2	DISCHARGED	Disjoint-support action cubes, selected cable counts, and the Lean representable HH_0 reduction.
C3	UNUSED	B -external naturality of C is a firewall; S uses Lemma 11.3, not C3.
P2/E7	UNUSED	Closed HJ Theorem 5.3 is not used as “ B is fixed”; S uses the reversed belt-sphere picture and Lemma 11.1.
P2/E10/S	DISCHARGED	Reversed 1-handle belt spheres miss the P0 cube and the C1 leftover link; chart returns miss every belt cube; dual loops pair to the identity; $b = 0$ foams.
P3/E11	DISCHARGED	Three 1–3 cancellations and a PL 4-ball; MWW Proposition 3.4 on X_J , not Σ_A^0 .
P3/E12	DISCHARGED	Computed: empty-link rank over $\mathbb{Q}$, PL I^4 with $\chi = 1$, glue $\chi = 2$, degree 494 (<code>scripts/certify_t73_e12_s4.py</code>). Cited: MWW Proposition 3.4 and Corollary 3.5 [17], arXiv:2206.04616; not proved here. Not X_J . Candidate <code>ExternalGeometry</code> uninhabited.
P3/E13	DISCHARGED	Constructed PL ψ , railroad attaching link, reduced PD, and Kirby pipeline; $X_J \cong \Sigma_A^0$. Lean <code>CSTopologyData</code> uninhabited.

The historical availability script reports the large files as absent. They are not used. The paper lemmas above do not supply a replacement proof.

The machine-readable allocation in `audit/t73_premise_audit.json` is regenerated by `scripts/audit_t73_premises.py`. It records generator-internal statuses separately from candidate-level closure. The C entry depends on Lemma 10.1 and equations (24)–(27), while S depends on Lemmas 11.1 and 11.3.

15. RETIRED ASSUMPTION REGISTER OF THE SEPTEMBER DRAFT

Earlier versions of this project collected unresolved candidate-specific geometry into five numbered assumptions A1–A5 in `paper/t73_candidate.tex`. That register is retained here only as a historical crosswalk. The controlling interface of the present paper is Hypotheses 3.2–3.5 together with the uninhabited Lean fields `ExternalGeometry` and `CSExternalGeometry`. None of A1–A5 is proved as stated, and none is used as a premise of Theorem 3.6.

Old assumption	Replacement	Status in this paper
A1 actual presentation and collar	P0a–c, P0d linking, P3/E13	DISCHARGED for the Johnson replacement; no historical-PD identity
A2 Hattori, endpoint, filtration, grading	C1, C2, Lemma 10.1	DISCHARGED for the Johnson replacement collar
A3 all-state two-handle cocone	C2, equations (26)–(27)	DISCHARGED for the Johnson replacement collar
A4 actual spheres and whole-source maps	S, Lemmas 11.1–11.3	DISCHARGED for the reversed belt-sphere picture
A5 rational closure and S^4 comparison	P3/E12 computed ranks; MWW 3.4–3.5 cited	Computed part DISCHARGED; citations not proved

What A1–A5 asked. A1 required the frozen framed diagram to be the actual reduced handle presentation of Σ_A^0 , with one common framed movie transporting collar, detector insertion and point-push chronology. A2 required the MWW one-handle hypotheses, endpoint shadow $u = e_2 - e_{87}$, whole-module filtration, and the four grading contributions $-44, +227, +315, -4$. A3 required the statewise divided-cubic beta/psi cocone on every finite cable state. A4 required embedded chosen spheres satisfying Horvat–Jabłonowski E7 and MWW three-handle formulas, with leading Frobenius form on the whole source. A5 required rational graded closure through the four-handle and a vanishing degree-494 piece for S^4 .

Why the register is not deleted. The old wording remains the most explicit statement of what would have been required to close the historical-PD route. Removing it would erase the claim boundary between finite certificates and geometric identification. Independent review of the retired route should still treat A1 as falsified for the unavailable planar diagram and A4 as the last load-bearing point of that manuscript; the Johnson replacement argument above supersedes that route without proving the old assumption statements verbatim.

How each old assumption could still be attacked.

- **A1.** Give a framed Kirby movie from the standard Cappell–Shaneson surgery to the frozen cut presentation, transporting every label, normal field, basepoint and detector insertion disk. The Johnson certificate replaces this for X_J , not for the historical PD.
- **A2.** Construct the actual coefficient equivalence from paired annuli and prove both action-naturality squares. Lemma 10.1 does this for the Johnson replacement collar only.
- **A3.** Prove the beta/psi equations on every finite cable state as equalities of linear maps on their full source spaces. Equations (26)–(27) do this after C1.
- **A4.** Embed the finite sphere models in one actual ∂W_2 , verify E7, and identify whole-source MWW maps with the leading Frobenius form. The reversed belt-sphere proof of S replaces the TH1/TH2/THXY combinatorial models.
- **A5.** Match published coefficient and grading conventions and check bidegree $(0, 0)$ for the four-handle isomorphism. E12 separates the computed ranks from the cited MWW isomorphisms.

Remark 15.1 (No inference from filenames or certificates). Names such as “global descending certificate”, “chosen sphere receipt”, or “Hattori decision” are locators for data. They do not occur as premises of any proof in this article. Whenever a finite file is used, the relevant mathematical content is stated explicitly and the replay command is given in Section G.

16. RELATED WORK

The Cappell–Shaneson literature contains both the original construction [5, 1] and substantial standardness results [2, 9, 14, 12]. Our use of Iwaki is limited to the standard matrix form, the representative table, and the homotopy-sphere criterion. His table entry for $(41, 189, 73)$ is motivation, not a singularity or exoticness theorem.

Manolescu–Neithalath and Manolescu–Walker–Wedrich provide the relevant handle calculus for skein lasagna modules [16, 17]. BPW quantum trace functoriality and the strict BHPW tangle theory provide general comparison tools [4, 3]. Section 7 applies these functors to the actual replacement coefficient and proves the statewise naturality.

Ren and Willis use Khovanov-theoretic skein lasagna methods to distinguish other exotic four-manifolds [19]. Their computations are independent of the trace-73 construction and are not used as its comparison theorem.

Horvat–Jabłonowski’s recent work gives a useful geometric criterion for recognizing three-handle attaching systems [11]. Theorem 5.3 is used only with Lemma 11.1 for kernel invariance in the closed manifold, not to fix B . Lemmas 5.5 and 5.7 do not appear in that paper and are not invoked.

17. REPRODUCIBILITY AND REVIEW

The proved finite and formal layers can be replayed in the following order.

- (1) Regenerate the public AR product witness, point-push, Hattori and owner-lift ledgers.
- (2) Verify the full 88-dimensional order-three matrix statement and its mutation controls.
- (3) Compile the Lean cubic, transported-pairing, Frobenius and quotient lemmas.
- (4) Regenerate the Johnson-bound C witness and check the statewise beta/psi formulas against MWW Definition 3.1.
- (5) Compile the paper and compare the field-by-field allocation with `T73External.lean`.
- (6) Regenerate `audit/t73_premise_audit.json` and verify its dependency on the actual maps and relative-sphere arguments in Sections 7–8.

The historical large artifacts are unavailable for this replay. Independent review should treat Lemmas 11.1–11.3 as discharged for the Johnson replacement reversed picture, Hypothesis 3.5 as discharged for the four-handle picture X_J , and the identification $X_J \cong \Sigma_A^0$ as the constructed CS handle picture.

18. CONCLUSION

P0, C, S, the MWW four-handle layer and the E13 CS handle picture are discharged for the explicit Johnson replacement. Lean `ExternalGeometry`

remains uninhabited, so Theorem 3.1 and Lean’s `computedCubic_eq_2624` still do not produce a counterexample.

APPENDIX A. RETIRED HISTORICAL OBJECTS

The following table records historical artifacts cited by earlier project notes. They are unavailable and provide no current proof input. The current argument does not use these historical objects. The digests are retained for provenance.

Priority	Object	SHA-256
P0	<code>cs_presentation.py</code>	B20BDDFF6FBF2A0E629083E2E62DD1DDF 9172A4E8893311CE122FFC4FEB388DFA
P0	<code>t73_eps0.erkmo.json</code>	6DF4D7906ADB18BA2D3237D4B49F8985 A5B18D8C39FF522AD6498F2369803FCB
P0	<code>t73_reduced_billiard.pd.json</code>	E6912A64457557469E5C691B4D57ABDB BF4C45ADB05492777C574223D0C06F8A
P0	<code>CUT_OBJECT.json</code>	4E1C4564ED98F30EAEBOA1AEAAD95ACA B5AD7B226C79B7D5BF3F259E8B56F541
P0	<code>BASE_INTERACTION_SUPPORT.json</code>	2D9E4760F836D9E04E6E0024CAE03A3D BB49A74FF5D129756F25450B30F10F7C
P0	<code>NORMAL_FIELD_MOVIE_CERT.json</code>	B2E97EB0981036768B8F1C1D5EF16F8 817EC1A7B951B6B32DBA2F9211FFD725A
P0	<code>T73_COLLAR_BRAID.json</code>	13AED414447ED33057C99D340910B501 09983CF36874C14ECE0E6A2834A12C24
P0	<code>verify_t73_collar_braid.py</code>	9D93A6CB320200A906F461EF3131127E A40C80A9A7828128E6F2804808417056
P1	<code>ACTUAL_PD_CABLE_UNIT_CERT.json</code>	7F3D3618D6A790A9B60EE8085B647AC2 AB742E1BC9C15841F1BEF015034217B5
P1	<code>PRODUCT_NORMAL_CHRISTOFFEL_THXY_MOVIE.json</code>	DE1AC479699EC79DE76D4265993DE437 493A3AAA6CABB636F98998644BF3181C
P2	<code>TH1_EXHAUSTIVE_ALL_ROW_GEOMETRY.json</code>	EE620E6B085A5F9E1C73CFDD1AD04FC 0682CEC74DA3DBF8AFE70DD19C038E3A0
P2	<code>TH2_EXHAUSTIVE_ALL_ROW_GEOMETRY.json</code>	4D1B627C0343A1C464319704EAFADCA1 27902A2E4E90CF1C283004359B7ADC24
P2	<code>THXY_FULL_MACRO_P3FREE_HJ_CERT.json</code>	EABF67C0D1AE309F0281297710A283B8 ED5A11D88C8E810837932313F4C53227

The same values and candidate paths are machine-readable in `audit/geometric_evidence_manifest.json`. These objects are not included. They cannot replace P0a–c, C1–C2 or S: a proof of P0 must construct the actual embedded detector and framed handle presentation; a proof of C must identify the actual coefficient/Hattori maps and their functorial role; and a proof of S must identify the actual essential-sphere endpoint maps with the MWW coequalizer maps.

APPENDIX B. APPENDIX B: STRICT P0 RECONSTRUCTION PROTOCOL

The repository now contains a strict, machine-checkable input contract in `audit/t73_p0_reconstruction_schema.json` and the verifier `scripts/reconstruct_t73_p0.py`. This protocol separates the public target braid from the AR geometry. The target word is regenerated from the primitive six-sweep rows; it is never accepted as input for constructing the AR collar.

Given a candidate with explicit strand coordinates, the command-line option `-derive-events` enumerates pairwise segment intersections in exact rational arithmetic and emits the elementary crossing movie. This mode does not read the public crossing rows; the emitted movie must subsequently pass the AR passage-binding and independent embeddedness gates.

The upstream reconstruction fixes a period-four scaled Freudenthal triangulation of the AR coordinate torus, factors A into 19 elementary integer moves, and pairs every move on H_B with the inverse-transpose move on H_D . The finite verifier checks after each step that the primal/dual intersection pairing remains the identity. Exact rational local templates cover all 52 unit slides, and 62 sequential support boxes have been placed in the fundamental cube away from the section arc. Exact grid routes connect the standard charts to every ordered pair of handle feet, and radius-1/32 tubes around each target/source pair are disjoint. This retired Nielsen branch did not give ψ_A : its local charts and thickened routes were not composed into one simplex-by-simplex homeomorphism. Consequently that branch made `scripts/check_t73_p0_pipeline.py` return `OVERALL=OPEN`. The Johnson branch supersedes it and is recorded in the committed passing certificate below.

An admissible input consists of exact rational PL data for a triangulated three-ball B , its boundary triangles, and 44 labelled height-monotone polylines

$$c_i : [0, 1] \longrightarrow B, \quad 1 \leq i \leq 44,$$

with a nonzero normal vector at every vertex. It must also contain a component/segment map from each c_i to the actual reduced AR attaching link, an endpoint-order map, and a normal-field transport map. Finally it contains the two cancellation movies, with owner labels and normal fields transported at every local move, and an ordered list of crossing events.

The crossing movie contains a derivation receipt with the canonical digest of the ball, all 44 strand polylines and the AR passage-binding map. The verifier recomputes this digest before reading any event, so the public crossing rows cannot be attached to unrelated symbolic geometry.

For each elementary crossing event at height s , the input records the two participating strands, the over-strand, the source segments, the sign and the Artin letter. The verifier derives the adjacent Artin generator from the current strand order and checks that the concatenated ordered word is exactly the independently regenerated public word of length 11340. The 252 public rows are used only to regenerate that target. Thus the final comparison is an equality of relative-endpoint braid words, not of exponent sums, writhe, permutation or hash.

The mathematical implication used by the protocol is the following.

Lemma B.1 (Soundness of the P0 certificate). *Suppose an input passes the strict verifier, and suppose its independent embeddedness, disjointness and local-movie receipts are valid for the stated PL complex. Then B is an embedded framed three-ball in the reduced AR handlebody, its 44 strands form a framed collar, and their relative-endpoint geometric braid equals the public 11340-letter pure braid.*

Proof. The PL ball and the passage-binding map place every c_i in the actual AR boundary chart and preserve ownership and endpoint order through the

two cancellations. Height monotonicity makes the c_i a braid in the collar; the disjointness receipt makes the strands pairwise disjoint. The normal vectors and their transport receipts give the framing on each strand and show that the two cancellation movies preserve it. Reading the certified crossing events in increasing height gives the relative-endpoint Artin word of this embedded tangle. The verifier's final equality identifies that word letter-for-letter with the public word. Since relative-endpoint isotopy classes of labelled braids are represented by their Artin words, the claimed braid equality follows. No step uses a word ledger to infer embeddedness. $\square$

The older committed object `audit/t73_ar_product_witness.json` does not satisfy this input contract: its parametrizations, normal fields and passage binding are descriptive strings, and its public crossing rows are imported from the frozen target input. Consequently the command

```
python3 scripts/reconstruct_t73_p0.py audit/t73_ar_product_witness.json
```

returns `P0_RECONSTRUCTION=OPEN`. The Johnson certificate `audit/t73\_p0\_johnson\_certificate.json` records `P0_STATUS=PROVED_FOR_EXPLICIT_JOHNSON_REPLACEMENT_PR`. The independent receipts named in `audit/t73\_p0\_reconstruction\_schema.json` apply to that Johnson reconstruction, not to the older symbolic witness.

APPENDIX C. THE COMPLETE DEGREE-THREE ENDPOINT VECTOR

With the conventions of Section 6, the coefficient vector $[\varepsilon^3](\rho_t(B_{88}) - I)u = (a_0, \dots, a_{87})^T$ is

$$\begin{aligned}
 (a_0, \dots, a_{87}) = & (-7052, -7052, 164, 164, 332, 332, 324, 324, 316, 316, 308, 308, \\
 & 300, 300, 292, 292, 284, 284, 276, 276, 268, 268, 260, 260, \\
 & 252, 252, 244, 244, 236, 236, 228, 228, 220, 220, 212, 212, \\
 & 204, 204, 196, 196, 188, 188, 180, 180, 172, 172, 164, 164, \\
 & 156, 156, 148, 148, 140, 140, 132, 132, 124, 124, 116, 116, \\
 & 108, 108, 100, 100, 92, 92, 84, 84, 76, 76, 68, 68, \\
 & 60, 60, 52, 52, 44, 44, 36, 36, 28, 28, 20, 20, \\
 (33) \quad & 12, 12, -164, -164).
 \end{aligned}$$

The detector covector is $\ell = e_{87}^* - e_2^*$, so $\ell(a_0, \dots, a_{87})^T = a_{87} - a_2 = -164 - 164 = -328$. The sum of the 88 entries is zero, as expected for the unreduced Burau difference on the invariant total-coordinate functional.

APPENDIX D. THE FOUR GRADING CONTRIBUTIONS

term	source	calculation
-44	remove the $p(N - 1)$ Hom normalization for $p = 44$, $N = 2$	$-44(2 - 1) = -44$
+227	227 distinct Frobenius labels, each of degree +1	$227 \cdot 1 = 227$
+315	one-handle gluing shift from 44 y -pairs and $271 = 44 + 227$ z -pairs	$44 + 271 = 315$
-4	cabled shift for $N = 2$, $\alpha = 0$, $ r = 2$	$(1 - 2)(2 \cdot 2 + 0) = -4$
total		$-44 + 227 + 315 - 4 = 494$

APPENDIX E. CONVENTION-LEGALITY CRITERIA

A serialization is legal only if it describes the same framed handle presentation, the same oriented based cut tangle, the same labelled boundary pairing, the same normal field, and the same insertion point. Its event order must be a linear extension of the geometric partial order. Adjacent events may be exchanged only when their supports are disjoint and a strict interchange theorem applies. Word replacement requires equality in the based labelled braid or tangle category; equality of closures, permutations, free-group words, or homology classes is insufficient.

A coordinate change P must act simultaneously:

$$(34) \quad W' = PWP^{-1}, \quad u' = Pu, \quad \ell' = \ell P^{-1},$$

with the shadow, collar, basepoint, and insertion point transported in the same naturality square. Under (34), $\ell'(W' - I)u' = \ell(W - I)u$. Thus coordinate invariance is proved when the simultaneous transport exists; it is not imposed by selecting the preferred numerical answer.

A collar is legal only if it is an actual framed tubular collar of the fixed component and is isotopic to the registered collar relative to the boundary, basepoints, labels, old input balls, normal framing, and insertion point. These criteria were written before the detailed alternative collar outputs were opened. Their use in the Johnson replacement is discharged by P0 and `scripts/reconstruct_t73_p0.py`, not by the unavailable historical PD.

APPENDIX F. TERMINOLOGY

Actual map: A map induced by a specified geometric tangle or surface cobordism, as opposed to an abstract linear map having the desired source and target.

Candidate-specific binding: An identification between immutable finite data and the actual geometric object required by a published theorem.

Complete quotient: The quotient by all relation generators in the applicable MWW handle formula, not by a sampled finite subset.

Divided cubic: The reduction modulo h of a map known on its whole domain to be divisible by h^3 , as in (18).

Endpoint shadow: The image of a coefficient trace class under the quantum vertical-to-horizontal trace and the fixed endpoint functor.

Finite certificate: Immutable finite data plus a deterministic exact checker. The term has no geometric force beyond the predicates actually recomputed.

Registered movie order: The oriented point-push chronology fixed in the public primitive input.

Johnson replacement: The splitting-preserving α_{ij} lift, 93-bit side choice, and CS handle picture discharged by P0–P3/E13 in this paper.

APPENDIX G. EXTENDED REPRODUCTION COMMANDS

The public repository is

<https://github.com/toffee-desuwa/smooth4pc-t73-lean>.

The commands below assume a fresh clone and Python 3.10 or later.

G.1. Reconstruct the detector.

```
python -I -B scripts/recompute_t73_delta3.py --check
```

Expected bearing lines include B44_LENGTH=11340, DELTA3_ETA_T1=2624, DELTA3_XI=0, and VERIFY=PASS.

G.2. Replay Johnson P0, C, S, and P3 certificates.

```
python3 scripts/certify_t73_p0_johnson.py --check
python3 scripts/certify_t73_c1_cut_link.py --check
python3 scripts/certify_t73_c2_comparison.py --check
python3 scripts/certify_t73_s_relative_moves.py --check
python3 scripts/certify_t73_p3_four_handle.py --check
python -B scripts/certify_t73_e12_s4.py --check
python -B scripts/certify_t73_e13_close.py --check
python3 scripts/audit_t73_premises.py --check
python -B scripts/check_t73_claim_boundary.py
```

G.3. Compile and inspect the Lean development. After installing `elan`, run

```
lake update
lake exe cache get
python -B tests/test_t73_minimal_formalization.py -v
lake lean T73Audit.lean
```

The Python test compiles the audited modules into a fresh temporary output root, requires exactly 38 axiom reports, rejects any axiom outside `propext`, `Classical.choice`, and `Quot.sound`, and rejects `sorryAx`.

G.4. Build this paper. From `paper/spc4-t73-candidate`:

```
pdflatex -interaction=nonstopmode -halt-on-error main.tex
bibtex main
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

The reviewed PDF is copied to `output/pdf/spc4-t73-candidate.pdf`. The retired manuscript `paper/t73_candidate.tex` remains in the repository as a historical ledger only; it is not the controlling document.

APPENDIX H. CORRESPONDENCE WITH THE LEAN FIELDS

For readers checking the formal boundary, the principal allocation is:

Lean field	Mathematical obligation
<code>x0, ell10, ell10_x0</code>	P1 comparison and the selected genuine MWW class.
<code>q01, ell11_comp_q01</code>	Complete two-handle beta/psi quotient and descent.
<code>q12, ell12_comp_q12</code>	Reversed belt spheres of S , MWW hemisphere maps, and complete coequalizer. Not the unused E7 “ B is fixed” claim.
<code>transport, fourIso</code>	E11 for the Johnson replacement X_J , not Σ_A^0 .
<code>s4DegreeZero</code>	Computed <code>EmptyKhQ(494)=0</code> and PL S^4 Euler data from <code>certify_t73_e12_s4.py</code> . The isomorphisms $G(B^4) \simeq G(S^4)$ and $\mathcal{S}_0^2(S^4) \cong \mathbb{Z}$ are MWW Proposition 3.4 and Corollary 3.5, cited not formalized. Empty-link Lean inhabitant only.
<code>diffeomorphismEquiv</code>	Graded diffeomorphism invariance of lasagna modules.
<code>matrixConditionsToHomotopySphere</code>	E13 constructed CS handle picture; Lean <code>CSTopologyData</code> uninhabited.

C3 and P2/E7 are Unused rather than Open. Lean `ExternalGeometry` is uninhabited for the Johnson candidate. The empty-link control in `T73S4Inhabitant.lean` packages only `EmptyKhQ`. The Lean theorem consumes no stronger statement, and it records the external topology as structure fields rather than formalizing it internally.

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